On the Largest Size of $(s, s+3)$-Core Partitions with Distinct Parts

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Abstract. The study of simultaneous core partitions has attracted considerable attention over the past two decades. In this paper, we determine the largest size of $(s, s+3)$-core partitions with distinct parts for $s \equiv 1$ or $2 \pmod 3$, and we show that the number of such partitions attaining the largest size is always 1. In particular, our result for $s \equiv 1 \pmod 3$ confirms a conjecture of Straub.

Keywords. Integer partition, Hook length, Core partition, Largest size.

MSC(2010). 05A17, 11P81.

1 Introduction

A partition λ is a finite weakly decreasing sequence of positive integers, denoted $\lambda = (\lambda_1, \lambda_2, \ldots, \lambda_\ell)$. The size of λ is $|\lambda| = \sum_{1 \le i \le \ell} \lambda_i$. By convention, the empty partition $\varnothing$ has size 0. A partition λ can be represented by its *Young diagram*, which is a left-justified array of boxes, with the i-th row containing λ_i boxes. For each box $\square$ in λ, its *hook* consists of the box $\square$ itself together with all boxes directly to its right and below it. The *hook length* of the box $\square$, denoted by $h_\square$, is the number of boxes in its hook. For example, the Young diagram and the hook lengths of the partition $(6, 3, 1)$ are given in Fig. 1.

An important notion in representation theory is the *t-core partition*, defined as a partition containing no hook lengths divisible by t. For example, consider the partition illustrated in Fig. 1. It is a 7-core partition since none of its hook lengths are divisible by 7. Let t_i $(1 \le i \le k)$ be positive integers. We say that λ is a $(t_1, t_2, \ldots, t_k)$-core partition if it is simultaneously a t_1-core, a t_2-core, ..., and a t_k-core partition.

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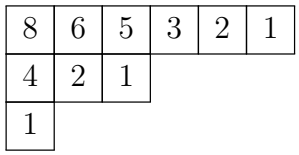


Fig. 1: The Young diagram and the hook lengths of the partition $(6, 3, 1)$.

Since Anderson's seminal work [3], the study of simultaneous core partitions has attracted considerable attention over the past two decades (see [2–4, 6–15, 17, 18, 21, 22, 24–28, 31–34]). Research has particularly focused on various statistical properties of these partitions, including their number, average size, and largest size. In this paper, we focus on determining the largest size of simultaneous core partitions. This topic has been extensively investigated, yielding several important results in the literature. For example, Olsson and Stanton [19] proved in 2007 that the largest size of an (s, t)-core partition is equal to $\frac{(s^2-1)(t^2-1)}{24}$, where s and t are coprime. Moreover, a partition attaining this largest size is self-conjugate. Nam and Yu [16] investigated the parity of parts in partitions. Specifically, they derived explicit formulas for the largest size of $(s, s+1)$-core partitions with all parts either odd or even. Following significant developments in the study of (s, t)-core partitions, considerable research attention has been paid to the investigation of simultaneous core partitions involving three or more distinct parameters. Yang, Zhong, and Zhou [30] established exact results for the largest size of $(t, t+1, t+2)$-core partitions. Subsequently, Xiong [25] generalized these findings by deriving a closed-form expression for the largest size of $(t, t+1, \ldots, t+p)$-core partitions, valid for arbitrary positive integers t and p. Related results are also given in Nath and Sellers [18], Baek, Nam, and Yu [5], and Sha and Xiong [23].

A burgeoning body of research on simultaneous core partitions with distinct parts originates from Amdeberhan's conjecture [1], which posits that the number of $(s, s+1)$-core partitions with distinct parts equals the Fibonacci number F_{s+1}. This conjecture has been proven and extended by many mathematicians, including Straub [22], Nath and Sellers [18], Zaleski and Zeilberger [32, 34], Xiong [26], and Paramonov [20], among others. In 2016, Straub [22] proposed several conjectures concerning the number and the largest size of $(s, s+2)$-core partitions with distinct parts. Specifically, for odd s, he conjectured that the largest size of such partitions is given by $\frac{1}{384}(s^2-1)(s+3)(5s+17)$. These conjectures were later proven by Yan, Qin, Jin, and Zhou [29], and further extended by Baek, Nam, and Yu [16], Paramonov [20], and other researchers. In 2017, Zaleski and Zeilberger [34] obtained explicit formulas for the moments of the sizes of these partitions. In 2016, Straub, in a private communication, conjectured an explicit formula for the largest size of $(s, s+3)$-core partitions with distinct parts, where $s \equiv 1 \pmod 3$.

In this paper, we prove the following results on the largest size of $(3t-2, 3t+1)$-core partitions with distinct parts. This confirms Straub's conjecture.

Theorem 1.1 (Straub's Conjecture) *For a positive integer t, the largest size of $(3t-2, 3t+1)$-core partitions with distinct parts is given by $\frac{1}{24}t(t^2-1)(9t+10)$. Moreover, the number of such partitions attaining the largest size is always 1.*

Building on Theorem 1.1, we further investigate the largest size of $(s, s+3)$-core partitions with distinct parts where $s \equiv 2 \pmod 3$. This leads to the following theorem.

Theorem 1.2 *For a positive integer t, the largest size of $(3t-1, 3t+2)$-core partitions with distinct parts is given by $\frac{1}{24}t(9t^3+38t^2+39t-14)$. Moreover, the number of such partitions attaining the largest size always equals 1.*

2 Preliminaries

In this section, we recall some basic notions and introduce the definition and preliminary results concerning β-sets of core partitions.

For a partition λ, the set of hook lengths of boxes in the first column of the Young diagram of λ is called the β-set of λ, denoted by $\beta(\lambda)$. In other words, the β-set of a partition $\lambda = (\lambda_1, \lambda_2, \ldots, \lambda_\ell)$ is given by

$$\beta(\lambda) = \{\lambda_i + \ell - i \mid 1 \le i \le \ell\}.$$

From Fig. 1, the hook lengths of the boxes in the first column of the partition $\lambda = (6,3,1)$ are 8, 4, and 1, giving $\beta(\lambda) = \{8,4,1\}$. Indeed, a partition is uniquely determined by its β-set. Let us review the following well-known results.

Lemma 2.1 ([3, 19, 25–27])

(1) The size of a partition λ is determined by its β-set as follows:

$$|\lambda| = \sum_{x \in \beta(\lambda)} x - \binom{|\beta(\lambda)|}{2},$$

where $|\beta(\lambda)|$ denotes the cardinality of the set $\beta(\lambda)$.

*(2) **(Abacus condition for t-core partitions)** A partition λ is a t-core partition if and only if for any $x \in \beta(\lambda)$ with $x \ge t$, we have $x - t \in \beta(\lambda)$.*

The following lemma characterizes the β-sets of partitions with distinct parts.

Lemma 2.2 ([26]) *A partition λ is a partition with distinct parts if and only if there exist no $x, y \in \beta(\lambda)$ with $x - y = 1$.*

Let P be a poset with a partial order $\prec$. For two elements $x, y \in P$, we say that y covers x if $x \prec y$ and no $z \in P$ satisfies $x \prec z \prec y$. We write $x \preceq y$ if $x \prec y$ or $x = y$. The Hasse diagram of a finite poset P is a graph in which the vertices represent the elements of P, and the edges represent the cover relations, where an edge connects x to y if and only if y covers x. In this case, y is drawn above x in the diagram. An order ideal of P is a subset I such that if $y \in I$ and $x \prec y$, then $x \in I$. We denote by $\mathcal{O}(P)$ the set of all order ideals of P.

For integers $a \leq b$, we define $[a, b] = \{a, a+1, \ldots, b\}$. Let $\mathbb{N}$ denote the set of all nonnegative integers. For coprime positive integers s and t, define the poset

$$P_{s,t} = \mathbb{N} \backslash \{\, n \in \mathbb{N} \mid n = k_1 s + k_2 t \text{ for some } k_1, k_2 \in \mathbb{N}\},$$

where the partial order is defined so that $x \in P_{s,t}$ covers $y \in P_{s,t}$ if $x - y = s$ or $x - y = t$. Using Lemma 2.1(2), Anderson [3] showed that the β-sets of (s, t)-core partitions are precisely the order ideals of $P_{s,t}$ as follows.

Lemma 2.3 ([3]) *Let s and t be coprime positive integers, and λ be a partition. Then λ is an (s, t)-core partition if and only if $\beta(\lambda)$ is an order ideal of $P_{s,t}$.*

By Lemma 2.2, an order ideal I of P is called *nice* if it contains no two elements $x, y \in I$ such that $x - y = 1$. We denote by $\mathcal{N}(P)$ the set of all nice order ideals of the poset P. Combining Lemmas 2.2 and 2.3, Yan, Qin, Jin, and Zhou [29] established the following result.

Lemma 2.4 ([29]) *Let s and t be coprime positive integers, and λ be a partition. Then λ is an (s, t)-core partition with distinct parts if and only if $\beta(\lambda)$ is a nice order ideal of $P_{s,t}$.*

Lemma 2.4 establishes a correspondence between $(3t-2, 3t+1)$-core partitions with distinct parts and the nice order ideals of $P_{3t-2,3t+1}$. The poset $P_{3t-2,3t+1}$ can be described as follows.

Lemma 2.5 *Let $t \geq 2$ be an integer. Then*

$$P_{3t-2,3t+1} = \bigcup_{s=0}^{3t-2} A_s,$$

where $A_0 = \{3i-2 \mid 1 \le i \le t-1\}$, $A_1 = \{3i-1 \mid 1 \le i \le 2t-2\}$, *and* $A_s = \{3i+(s-2)(3t+1) \mid 1 \le i \le 3t-1-s\}$ *for* $2 \le s \le 3t-2$.

Proof. First, we show that $\bigcup_{s=0}^{3t-2} A_s \subseteq P_{3t-2,3t+1}$. By the definition of $P_{3t-2,3t+1}$, it is easy to check that A_0 and A_1 are contained in $P_{3t-2,3t+1}$. Thus, it remains to prove $\bigcup_{s=2}^{3t-2} A_s \subseteq P_{3t-2,3t+1}$. Assume, for contradiction, that there exists an element $y \in \bigcup_{s=2}^{3t-2} A_s$ such that $y \notin P_{3t-2,3t+1}$. Choose y to be the smallest such element. By the definition of A_s, we have $y = 3i+(s-2)(3t+1)$ for some $1 \le i \le 3t-1-s$ and $2 \le s \le 3t-2$. Since $y \notin P_{3t-2,3t+1}$, it follows that y can be written as $y = k_1(3t-2) + k_2(3t+1)$ for some nonnegative integers k_1 and k_2. We claim that $k_2 = 0$. If not, we have $k_2 > 0$, and hence $s > 2$. Note that $y - (3t+1) = 3i + (s-3)(3t+1) \in \bigcup_{s=2}^{3t-2} A_s$, but this element is not in $P_{3t-2,3t+1}$, which contradicts the minimality of y. Hence $k_2 = 0$, and $y = 3i + (s-2)(3t+1) = k_1(3t-2)$. Therefore, $3i + 3s - 6 = (k_1 - s + 2)(3t-2)$, which implies that $k_1 - s + 2$ is a positive multiple of 3. Hence $k_1 - s + 2 \ge 3$, and thus $k_1 \ge s+1$. Consequently, we have $(k_1 - s + 2)(3t-2) \ge 9t-6$. On the other hand, since $i \le 3t-1-s$, we have $3i+3s-6 \le 9t-9$. Comparing these two inequalities yields $3i + 3s - 6 < (k_1 - s + 2)(3t-2)$, which contradicts the previous equality. Hence, we have $\bigcup_{s=2}^{3t-2} A_s \subseteq P_{3t-2,3t+1}$. Therefore, recalling that $A_0 \cup A_1 \subseteq P_{3t-2,3t+1}$, we conclude that $\bigcup_{s=0}^{3t-2} A_s \subseteq P_{3t-2,3t+1}$.

It remains to verify the inclusion $P_{3t-2,3t+1} \subseteq \bigcup_{s=0}^{3t-2} A_s$ to complete the proof. Let y be an element of $P_{3t-2,3t+1}$. Then $y = m + k(3t+1)$ for some integers m and k with $1 \le m \le 3t$, $m \ne 3t-2$, and $k \ge 0$. To show that $y \in \bigcup_{s=0}^{3t-2} A_s$, we consider three cases.

Case 1. Suppose $m \equiv 0 \pmod 3$. Then $y = 3i+k(3t+1)$ for some $1 \le i \le t$ and $k \ge 0$. Let $p = k+2$. Then $y = 3i+(p-2)(3t+1)$ for some $1 \le i \le t$ and $p \ge 2$. Thus, it suffices to show that $i \le 3t-p-1$ and $p \le 3t-2$. Assume, for contradiction, that $i > 3t-p-1$, thus $p \ge 3t-i$. Then $y = 3i+(p-2)(3t+1) = (3t+1-i)(3t-2)+(p-3t+i)(3t+1) \notin P_{3t-2,3t+1}$, a contradiction. Thus, we have $i \le 3t-p-1$. Since $i \ge 1$, it follows that $p \le 3t-2$. Consequently, we conclude that $y \in \bigcup_{s=2}^{3t-2} A_s$.

Case 2. Suppose $m \equiv 1 \pmod 3$. Then $y = 3i-2+k(3t+1)$ for some $1 \le i \le t-1$ and $k \ge 0$.

- If $k = 0$, then $y = 3i-2$ for $1 \le i \le t-1$. By the definition of A_0, this implies $y \in A_0$.
- If $k = 1$, then $y = 3i-2+3t+1 = 3(t+i)-1$. To satisfy the definition of the poset, we must have $1 \le i \le t-2$. Indeed, if $i = t-1$, then $y = 3(t+t-1)-1 = 2(3t-2)$, which shows that $y \notin P_{3t-2,3t+1}$. Let $h = t+i$. Then $y = 3h-1$ for $t+1 \le h \le 2t-2$. By the definition of A_1, we have $y \in A_1$.
- If $k \ge 2$, then $y = 3i-2+(k-2)(3t+1)+2(3t+1) = 3(2t+i)+(k-2)(3t+1)$ for some $1 \le i \le t-1$. Let $h = 2t+i$. Then $y = 3h+(k-2)(3t+1)$ for some

$2t+1 \le h \le 3t-1$. Thus, it suffices to show that $h \le 3t-k-1$ and $k \le 3t-2$. Assume to the contrary that $k \ge 3t-h$. Then $y = 3h+(k-2)(3t+1) = (3t+1-h)(3t-2)+(k-3t+h)(3t+1) \notin P_{3t-2,3t+1}$, a contradiction. Hence, $h \le 3t-k-1$. Since $h \ge 2t+1$, it follows that $k \le t-2$. Note that this is stronger than the original bound $k \le 3t-2$. Therefore, $y \in \bigcup_{s=2}^{3t-2} A_s$.

Case 3. Suppose $m \equiv 2 \pmod 3$. Then $y = 3i-1+k(3t+1)$ for some $1 \le i \le t$ and $k \ge 0$.

- If $k = 0$, then $y = 3i-1$ for $1 \le i \le t$. By the definition of A_1, this implies $y \in A_1$.
- If $k \ge 1$, then $y = 3i-1+(k-1)(3t+1)+(3t+1) = 3(t+i)+(k-1)(3t+1)$ for some $1 \le i \le t$. Let $h = t+i$ and $p = k+1$. Then $y = 3h+(p-2)(3t+1)$ with $t+1 \le h \le 2t$ and $p \ge 2$. Thus, it suffices to show that $h \le 3t-p-1$ and $p \le 3t-2$. Assume to the contrary that $p \ge 3t-h$. Then $y = 3h+(p-2)(3t+1) = (3t+1-h)(3t-2)+(p-3t+h)(3t+1) \notin P_{3t-2,3t+1}$, a contradiction. Thus, $h \le 3t-p-1$. Combining this with $h \ge t+1$, we obtain $p \le 2t-2$, which is stronger than the original bound $p \le 3t-2$. Therefore, $y \in \bigcup_{s=2}^{3t-2} A_s$.

Combining the above three cases, we obtain $P_{3t-2,3t+1} \subseteq \bigcup_{s=0}^{3t-2} A_s$. This completes the proof. ∎

By Lemma 2.5, the Hasse diagram of the poset $P_{3t-2,3t+1}$ is described as follows.

(i) For $2 \le s \le 3t-3$, each element $x \in A_{s+1}$ covers exactly two elements in A_s, that is, $x-(3t+1)$ and $x-(3t-2)$.

(ii) For $1 \le i \le 2t-3$, each element $3t+3i \in A_2$ covers exactly two elements $3i-1$ and $3i+2$ in A_1. In addition, the element $3t \in A_2$ covers 2 in A_1.

(iii) For $1 \le i \le t-2$, each element $3t+3i-1 \in A_1$ covers exactly two elements $3i-2$ and $3i+1$ in A_0. In addition, the element $3t-1 \in A_1$ covers 1 in A_0.

As an example, Fig. 2 shows the Hasse diagram of the poset $P_{13,16}$. In the poset $P_{3t-2,3t+1}$, the elements of rank 0 are defined to be $1, 2, 3, \ldots, 3t-3$.

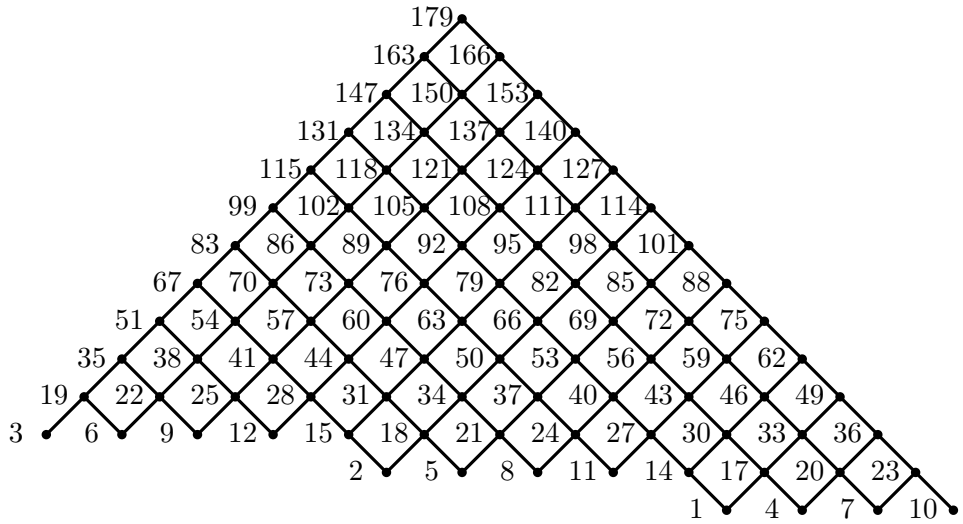


Fig. 2: The Hasse diagram of the poset $P_{13,16}$.

3 Proof of Theorem 1.1

This section investigates the largest size of $(3t-2, 3t+1)$-core partitions with distinct parts. To prove Theorem 1.1, we construct the unique $(3t-2, 3t+1)$-core partition into distinct parts that attains the largest size.

Denote by $\mathcal{W}_{3t-2,3t+1}$ the set of all $(3t-2, 3t+1)$-core partitions with distinct parts. By Lemma 2.4, we have $\beta(\lambda) \in \mathcal{N}(P_{3t-2,3t+1})$ if $\lambda \in \mathcal{W}_{3t-2,3t+1}$. Note that $3t-1$ and $3t$ cannot simultaneously belong to the set $\beta(\lambda)$. Following the definition of $\mathcal{N}(P_{3t-2,3t+1})$, we divide this set into three classes:

(i) $\mathcal{M}(P_{3t-2,3t+1})$, the set of all nice order ideals $I \in \mathcal{N}(P_{3t-2,3t+1})$ with $3t-1 \notin I$ and $3t \notin I$,

(ii) $\mathcal{T}(P_{3t-2,3t+1})$, the set of all nice order ideals $I \in \mathcal{N}(P_{3t-2,3t+1})$ with $3t-1 \in I$,

(iii) $\mathcal{S}(P_{3t-2,3t+1})$, the set of all nice order ideals $I \in \mathcal{N}(P_{3t-2,3t+1})$ with $3t \in I$.

Thus,

$$\mathcal{N}(P_{3t-2,3t+1}) = \mathcal{M}(P_{3t-2,3t+1}) \cup \mathcal{T}(P_{3t-2,3t+1}) \cup \mathcal{S}(P_{3t-2,3t+1}).$$

This decomposition naturally yields three cases for the analysis that follows.

3.1 The case $\beta(\lambda) \in \mathcal{M}(P_{3t-2,3t+1})$

By the definition of $\mathcal{M}(P_{3t-2,3t+1})$ and Lemma 2.5, one can directly deduce the following result.

Lemma 3.1 *Let t be an integer satisfying $t \geq 2$. For $1 \leq s \leq t-1$, define*

$$Q_s^{(1)} = \{3i - 2 + (s-1)(3t+1) \mid 1 \leq i \leq t-s\},$$

$$Q_s^{(2)} = \{3i - 1 + (s-1)(3t+1) \mid 1 \leq i \leq t-s\},$$

and

$$Q_s^{(3)} = \{3i + (s-1)(3t+1) \mid 1 \leq i \leq t-s\}.$$

Define the poset

$$B_{3t-2,3t+1} = (\bigcup_{s=1}^{t-1} Q_s^{(1)}) \cup (\bigcup_{s=1}^{t-1} Q_s^{(2)}) \cup (\bigcup_{s=1}^{t-1} Q_s^{(3)}).$$

Then we have

$$\mathcal{M}(P_{3t-2,3t+1}) = \mathcal{N}(B_{3t-2,3t+1}).$$

By Lemma 3.1, we construct the Hasse diagram of the poset $B_{3t-2,3t+1}$. Fig. 3 illustrates the Hasse diagram of the poset $B_{13,16}$. In Fig. 3, from right to left are $\bigcup_{s=1}^{4} Q_s^{(1)}$, $\bigcup_{s=1}^{4} Q_s^{(2)}$, and $\bigcup_{s=1}^{4} Q_s^{(3)}$, respectively.

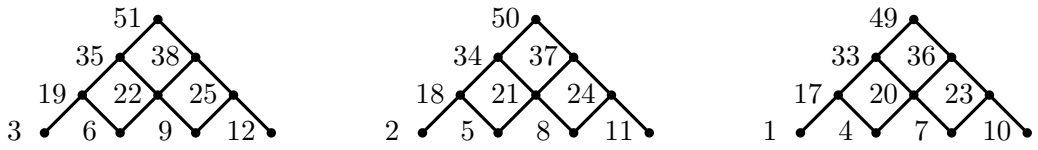


Fig. 3: The Hasse diagram of the poset $B_{13,16}$.

To characterize the nice order ideals in $\mathcal{N}(B_{3t-2,3t+1})$ that correspond to the $(3t-2, 3t+1)$-core partition into distinct parts with the largest size, with the β-set is contained in $\mathcal{M}(P_{3t-2,3t+1})$, we require the following three lemmas.

Lemma 3.2 *For an integer $t \geq 2$, let λ be a $(3t-2, 3t+1)$-core partition into distinct parts of the largest size with $\beta(\lambda) \in \mathcal{M}(P_{3t-2,3t+1})$. Suppose $\beta(\lambda)$ contains an element $x \in Q_s^{(1)}$ (resp. $x \in Q_s^{(2)}$, $x \in Q_s^{(3)}$) for $1 \leq s \leq t-1$. Then y must also belong to $\beta(\lambda)$ for all $y \in Q_s^{(1)}$ (resp. $y \in Q_s^{(2)}$, $y \in Q_s^{(3)}$) with $y > x$.*

Proof. Suppose that λ is a $(3t-2,3t+1)$-core partition into distinct parts of the largest size, and that $\beta(\lambda) \in \mathcal{M}(P_{3t-2,3t+1})$. To prove the lemma, it suffices to show that for every $1 \le s \le t-1$, if $\beta(\lambda)$ contains an element $x \in Q_s^{(1)}$ (resp. $x \in Q_s^{(2)}$, $x \in Q_s^{(3)}$), then it also contains $x+3 \in Q_s^{(1)}$ (resp. $x+3 \in Q_s^{(2)}$, $x+3 \in Q_s^{(3)}$). Assume, for contradiction, that this does not hold. Let s be the minimal index for which the property fails, and let x be the smallest such element in $Q_s^{(1)}$ (resp. $Q_s^{(2)}$, $Q_s^{(3)}$). If $s > 1$, then replace y by $y+3$ whenever $p \preceq y$ for some $p \le x$ and $p \in Q_s^{(1)}$ (resp. $p \in Q_s^{(2)}$, $p \in Q_s^{(3)}$). If $s = 1$, by the minimality of x, we have $\{1,4,\ldots,x\} \subseteq \beta(\lambda)$ (resp. $\{2,5,\ldots,x\} \subseteq \beta(\lambda)$, $\{3,6,\ldots,x\} \subseteq \beta(\lambda)$), which implies that $\{2,5,\ldots,x+1\} \cap \beta(\lambda) = \varnothing$ (resp. $\{3,6,\ldots,x+1\} \cap \beta(\lambda) = \varnothing$, $\{4,7,\ldots,x+1\} \cap \beta(\lambda) = \varnothing$). Then replace y by $y+1$ whenever $p \preceq y$ for some $p \le x$ and $p \in Q_1^{(1)}$ (resp. $p \in Q_1^{(2)}$, $p \in Q_1^{(3)}$). Moreover, for every $k \in \{2,5,\ldots,x+1\}$ (resp. $k \in \{3,6,\ldots,x+1\}$, $k \in \{4,7,\ldots,x+1\}$), we have $k \in Q_1^{(2)}$ (resp. $k \in Q_1^{(3)}$, $k \in Q_1^{(1)}$). Hence, this construction yields a new order ideal $\beta' \in \mathcal{M}(P_{3t-2,3t+1})$ with $|\beta'| = |\beta(\lambda)|$ and $\sum_{x\in\beta'} x > \sum_{x\in\beta(\lambda)} x$. By Lemma 2.1(1), the corresponding partition has strictly larger size than λ, contradicting the maximality of λ. This completes the proof. ∎

For an integer t, let U_t denote the unique order ideal consisting of all elements of $P_{t,t+1}$. For $1 \le i \le 3(t-1)$, let $\beta_{i,0}$ be the unique order ideal in $\mathcal{O}(B_{3t-2,3t+1})$ that contains the elements $i+3, i+6, \ldots, i+3(t-1-\lfloor \frac{i+2}{3} \rfloor)$ and is isomorphic to $U_{t-\lfloor \frac{i+2}{3} \rfloor}$. Note that $\beta_{i,0} = \varnothing$ when $i \in \{3(t-1)-2, 3(t-1)-1, 3(t-1)\}$. Moreover, one easily checks that each $\beta_{i,0}$ is a nice order ideal of $P_{3t-2,3t+1}$, i.e., $\beta_{i,0} \in \mathcal{N}(P_{3t-2,3t+1})$.

For $1 \le i \le 3(t-1)$ and $1 \le j \le t-\lfloor \frac{i+2}{3} \rfloor$, define $\beta_{i,j}$ to be the unique order ideal in $\mathcal{O}(B_{3t-2,3t+1})$, which contains $\beta_{i,0}$ and the chain consisting of $i, i+(3t+1), \ldots, i+(j-1)(3t+1)$. Note that $\beta_{3(t-1),1} = \{3(t-1)\}$, $\beta_{3(t-1)-1,1} = \{3(t-1)-1\}$, and $\beta_{3(t-1)-2,1} = \{3(t-1)-2\}$. Clearly, $\beta_{1,t-1}$ (resp. $\beta_{2,t-1}$, $\beta_{3,t-1}$) is exactly the set consisting of all elements of $\bigcup_{s=1}^{t-1} Q_s^{(1)}$ (resp. $\bigcup_{s=1}^{t-1} Q_s^{(2)}$, $\bigcup_{s=1}^{t-1} Q_s^{(3)}$). For example, as shown in Fig. 3, the order ideals $\beta_{1,4}$, $\beta_{2,4}$, and $\beta_{3,4}$ are displayed from right to left, respectively, with their elements represented by solid circles. Moreover, one easily checks that each $\beta_{i,j}$ is a nice order ideal of $P_{3t-2,3t+1}$, that is, $\beta_{i,j} \in \mathcal{N}(P_{3t-2,3t+1})$. Let $\gamma_{i,j}$ be the unique partition such that $\beta(\gamma_{i,j}) = \beta_{i,j}$. By Lemma 2.4, it follows that $\gamma_{i,j} \in \mathcal{W}_{3t-2,3t+1}$.

Lemma 3.3 *For an integer $t \ge 2$, let λ be a $(3t-2,3t+1)$-core partition into distinct parts with $\beta(\lambda) \in \mathcal{M}(P_{3t-2,3t+1})$. If λ has the largest size, then there exist some integers $1 \le i \le 3(t-1)$ and $1 \le j \le t-\lfloor \frac{i+2}{3} \rfloor$ such that $\beta(\lambda) = \beta_{i,j}$.*

Proof. Suppose that λ is a $(3t-2,3t+1)$-core partition into distinct parts of the largest size, and $\beta(\lambda) \in \mathcal{M}(P_{3t-2,3t+1})$. Let $i \in \beta(\lambda)$ be the smallest element of rank 0, with $1 \le i \le 3(t-1)$. Once i is fixed, let j be the largest integer such that

$i+(j-1)(3t+1) \in \beta(\lambda)$, where $1 \le j \le t - \lfloor \frac{i+2}{3} \rfloor$. To prove the lemma, it suffices to show that $i+3+m(3t+1) \in \beta(\lambda)$ for all $0 \le m \le t - \lfloor \frac{i+2}{3} \rfloor - 2$. Assume, for contradiction, that the lemma does not hold. Then there exists an integer ℓ such that $i+3+\ell(3t+1) \notin \beta(\lambda)$. By Lemma 3.2, we have $j \le \ell \le t - \lfloor \frac{i+2}{3} \rfloor - 2$. Define λ' to be a $(3t-2, 3t+1)$-core partition whose corresponding $\beta(\lambda')$ is obtained from $\beta(\lambda)$ by replacing $i+(j-1)(3t+1)$ with $i+3+\ell(3t+1)$. By Lemma 3.2 and the definition of $\mathcal{M}(P_{3t-2,3t+1})$, $\beta(\lambda')$ is a nice order ideal of $P_{3t-2,3t+1}$, and hence $\beta(\lambda') \in \mathcal{M}(P_{3t-2,3t+1})$. Moreover, one checks that $\beta(\lambda')$ has the same cardinality as $\beta(\lambda)$, while the sum of its elements is larger. By Lemma 2.1(1), it follows that the size of λ' is larger than the size of λ, which contradicts the assumption that λ has the largest size. This completes the proof. ∎

Lemma 3.4 *For $1 \le i \le 3(t-1)$, we have $|\gamma_{i,j}| \le |\gamma_{i,t-\lfloor \frac{i+2}{3} \rfloor}|$ for $1 \le j \le t - \lfloor \frac{i+2}{3} \rfloor$, with the equality holding if and only if $j = t - \lfloor \frac{i+2}{3} \rfloor$. Moreover, when $1 \le i \le 3(t-2)$ and $t \ge 3$, we have $|\gamma_{i,t-\lfloor \frac{i+2}{3} \rfloor}| > |\gamma_{i+3,t-\lfloor \frac{i+2}{3} \rfloor - 1}|$.*

Proof. By Lemma 2.1(1), we have

$$\begin{aligned}
|\gamma_{i,j}| &= \sum_{h \in \beta_{i,j}} h - \binom{|\beta_{i,j}|}{2} \\
&= \sum_{h \in \beta_{i,0}} h + \sum_{p=0}^{j-1} (i + p(3t+1)) - \binom{|\beta_{i,0}| + j}{2} \\
&= \sum_{h \in \beta_{i,0}} h - \binom{|\beta_{i,0}|}{2} + \sum_{p=0}^{j-1} (i + p(3t+1)) - \binom{|\beta_{i,0}| + j}{2} + \binom{|\beta_{i,0}|}{2} \\
&= |\gamma_{i,0}| + ij + 3t\binom{j}{2} - |\beta_{i,0}|j.
\end{aligned}$$

By the definition of $\beta_{i,0}$, we have $|\beta_{i,0}| = \sum_{j=1}^{t-\lfloor \frac{i+2}{3} \rfloor - 1} j = \binom{t-\lfloor \frac{i+2}{3} \rfloor}{2}$. Hence, we have

$$|\gamma_{i,j}| = |\gamma_{i,0}| + ij + 3t\binom{j}{2} - j\binom{t - \lfloor \frac{i+2}{3} \rfloor}{2}. \tag{3.1}$$

Furthermore, by setting $j = 1$ and $j = t - \lfloor \frac{i+2}{3} \rfloor$ in (3.1), we obtain the following result:

$$|\gamma_{i,t-\lfloor \frac{i+2}{3} \rfloor}| - |\gamma_{i,0}| = \begin{cases} 3m(t-m) + (2t+m)\binom{t-m}{2} & \text{if } i = 3m, \\ (3m-1)(t-m) + (2t+m)\binom{t-m}{2} & \text{if } i = 3m-1, \\ (3m-2)(t-m) + (2t+m)\binom{t-m}{2} & \text{if } i = 3m-2, \end{cases} \tag{3.2}$$

and

$$|\gamma_{i,1}| - |\gamma_{i,0}| = \begin{cases} 3m - \binom{t-m}{2} & \text{if } i = 3m, \\ 3m - 1 - \binom{t-m}{2} & \text{if } i = 3m-1, \\ 3m - 2 - \binom{t-m}{2} & \text{if } i = 3m-2, \end{cases} \tag{3.3}$$

for $1 \le m \le t-2$. Subtracting (3.3) from (3.2), we get

$$|\gamma_{i,t-\lfloor \frac{i+2}{3} \rfloor}| > |\gamma_{i,1}| \tag{3.4}$$

for $1 \le i \le 3(t-2)$. In particular, if $i = 3(t-1)$ (resp. $i = 3(t-1)-1$, $i = 3(t-1)-2$), then we derive that

$$|\gamma_{i,t-\lfloor \frac{i+2}{3} \rfloor}| = |\gamma_{i,1}|, \tag{3.5}$$

and its corresponding β-set is $\{3(t-1)\}$ (resp. $\{3(t-1)-1\}$, $\{3(t-1)-2\}$).

On the other hand, for fixed t and i, we observe that $|\gamma_{i,j}|$ is a quadratic function in j with a positive leading coefficient. Consequently, when j ranges over $[1, t-\lfloor \frac{i+2}{3} \rfloor]$, the maximum value of $|\gamma_{i,j}|$ is attained at one of the endpoints $j = 1$ or $j = t - \lfloor \frac{i+2}{3} \rfloor$. Combining (3.4) and (3.5), we conclude that $|\gamma_{i,j}| \le |\gamma_{i,t-\lfloor \frac{i+2}{3} \rfloor}|$ for $1 \le j \le t - \lfloor \frac{i+2}{3} \rfloor$, with equality if and only if $j = t - \lfloor \frac{i+2}{3} \rfloor$.

From the definitions of $\beta_{i-3,0}$ and $\beta_{i,j}$, we establish the key relation $\beta_{i,t-\lfloor \frac{i+2}{3} \rfloor} = \beta_{i-3,0}$ for all $4 \le i \le 3(t-1)$. Combining this relation with (3.2) yields the cardinality inequality $|\gamma_{i,t-\lfloor \frac{i+2}{3} \rfloor}| > |\gamma_{i,0}| = |\gamma_{i+3,t-\lfloor \frac{i+2}{3} \rfloor - 1}|$. This inequality holds for all $t \ge 3$ and $1 \le i \le 3(t-2)$, which completes the proof. ∎

The following theorem is the main result of this subsection.

Theorem 3.5 *For an integer $t \ge 2$, let λ be a $(3t-2, 3t+1)$-core partition into distinct parts with $\beta(\lambda) \in \mathcal{M}(P_{3t-2,3t+1})$. If λ has the largest size, then $\beta(\lambda) = \beta_{3,t-1}$. Moreover, the number of such partitions attaining the largest size is* 1.

Proof. Suppose that λ is a $(3t-2, 3t+1)$-core partition into distinct parts with the largest size and $\beta(\lambda) \in \mathcal{M}(P_{3t-2,3t+1})$. From Lemmas 3.3 and 3.4, we find that $\beta(\lambda)$ equals $\beta_{1,t-1}$, $\beta_{2,t-1}$, or $\beta_{3,t-1}$. By the definition of $\beta_{1,t-1}$, $\beta_{2,t-1}$, and $\beta_{3,t-1}$, we have

$$|\beta_{1,t-1}| = |\beta_{2,t-1}| = |\beta_{3,t-1}|,$$

$$\sum_{h \in \beta_{3,t-1}} h = \sum_{h \in \beta_{2,t-1}} h + \frac{t(t-1)}{2},$$

and

$$\sum_{h \in \beta_{2,t-1}} h = \sum_{h \in \beta_{1,t-1}} h + \frac{t(t-1)}{2}.$$

By Lemma 2.1(1) and combining the three equations above, we obtain that

$$|\gamma_{3,t-1}| - |\gamma_{2,t-1}| = \frac{t(t-1)}{2} \tag{3.6}$$

and

$$|\gamma_{2,t-1}| - |\gamma_{1,t-1}| = \frac{t(t-1)}{2}. \tag{3.7}$$

Adding (3.6) and (3.7), we have

$$|\gamma_{3,t-1}| - |\gamma_{1,t-1}| = t(t-1). \tag{3.8}$$

Hence, we have $|\gamma_{3,t-1}| > |\gamma_{2,t-1}| > |\gamma_{1,t-1}|$ for $t \geq 2$. It follows that $\lambda = \gamma_{3,t-1}$ and $\beta(\lambda) = \beta_{3,t-1}$ as desired. Moreover, one can easily check that the number of partitions of the largest size is 1, which completes the proof. ∎

3.2 The case $\beta(\lambda) \in \mathcal{T}(P_{3t-2,3t+1})$

The purpose of this subsection is to study the $(3t-2, 3t+1)$-core partition λ with distinct parts such that $\beta(\lambda) \in \mathcal{T}(P_{3t-2,3t+1})$. In what follows, we describe $\beta(\lambda)$ for λ of the largest size by the following results.

Lemma 3.6 *For an integer $t \geq 2$, let λ be a $(3t-2, 3t+1)$-core partition into distinct parts of the largest size. If $\beta(\lambda) \in \mathcal{T}(P_{3t-2,3t+1})$, then $\beta(\lambda)$ contains all elements of A_0.*

Proof. Recall that $A_0 = \{3i-2 \mid 1 \leq i \leq t-1\}$. Let λ be a $(3t-2, 3t+1)$-core partition into distinct parts of the largest size such that $\beta(\lambda) \in \mathcal{T}(P_{3t-2,3t+1})$. Suppose, for contradiction, that the lemma does not hold. Since $3t-1 \in \beta(\lambda)$, it follows from Lemma 2.1(2) that $1 \in \beta(\lambda)$. Let $3m-2$ be the smallest element of A_0 that is not contained in $\beta(\lambda)$, where $2 \leq m \leq t-1$. Once m is fixed, let ℓ be the largest integer, with $-1 \leq \ell \leq t-m-1$, such that the element $3(m+\ell)-1 \in A_1$ of rank 0 is not contained in $\beta(\lambda)$. We now categorize $\beta(\lambda)$ into three classes I_1, I_2, and I_3 as follows.

(i) $I_1 = \{1, 4, \ldots, 3(m-1)-2\}$.

(ii) $I_2 \in \mathcal{O}(C_{3t-2,3t+1})$, where the poset $C_{3t-2,3t+1}$ is given by:

$$C_{3t-2,3t+1} = (\bigcup_{s=1}^{\ell+2} K_s^{(1)}) \cup (\bigcup_{s=1}^{\ell} K_s^{(2)}) \cup (\bigcup_{s=1}^{\ell} K_s^{(3)}).$$

Here, the constituent sets are defined as

$$K_s^{(1)} = \{3i + (s-1)(3t+1) \mid m-1 \leq i \leq m+\ell+1-s\} \quad \text{for } 1 \leq s \leq \ell+2,$$
$$K_s^{(2)} = \{3i-1 + (s-1)(3t+1) \mid m \leq i \leq m+\ell-s\} \quad \text{for } 1 \leq s \leq \ell,$$
$$K_s^{(3)} = \{3i-2 + (s-1)(3t+1) \mid m+1 \leq i \leq m+\ell+1-s\} \quad \text{for } 1 \leq s \leq \ell.$$

(iii) $I_3 \in \mathcal{O}(D_{3t-2,3t+1})$, where $D_{3t-2,3t+1} = \bigcup_{s=1}^{t-\ell-2} K_s^{(4)}$ with

$$K_s^{(4)} = \{3i-1+(s-1)(3t+1) \mid m+\ell+1 \le i \le m+t-1-s\} \text{ for } 1 \le s \le t-\ell-2.$$

Note that $3t-1 \in I_3$.

According to whether $3(t-1)$ and $3(m-1)$ belong to $\beta(\lambda)$, we divide the analysis into the following four cases: (1) $3(t-1), 3(m-1) \notin \beta(\lambda)$, (2) $3(t-1) \notin \beta(\lambda)$ and $3(m-1) \in \beta(\lambda)$, (3) $3(t-1), 3(m-1) \in \beta(\lambda)$, (4) $3(t-1) \in \beta(\lambda)$ and $3(m-1) \notin \beta(\lambda)$. We prove the assertion only for Case (1). The remaining cases follow by similar arguments.

Case (1). $3(t-1), 3(m-1) \notin \beta(\lambda)$. We proceed to construct a new order ideal β' of $P_{3t-2,3t+1}$ using the following procedure.

- Replace I_1 by $\{1, 4, \ldots, 3(m-1)-2, 3m-2\}$.
- Replace each element x of I_2 by $x+1$.
- Replace each element x of I_3 by $x+3$.

Suppose that the cardinality of I_3 is given by y. Then

$$|\beta'| = |\beta(\lambda)| + 1$$

and

$$\sum_{h \in \beta'} h = \sum_{h \in \beta(\lambda)} h + (3m-2) + 3y + [|\beta(\lambda)| - y - (m-1)] = \sum_{h \in \beta(\lambda)} h + |\beta(\lambda)| + 2y + 2m - 1.$$

Let λ' be the unique partition such that $\beta(\lambda') = \beta'$. By Lemma 2.1(1), the size of λ' is given by

$$\begin{aligned} |\lambda'| &= \sum_{h \in \beta'} h - \binom{|\beta'|}{2} \\ &= \sum_{h \in \beta(\lambda)} h + |\beta(\lambda)| + 2y + 2m - 1 - \binom{|\beta(\lambda)|}{2} + \binom{|\beta(\lambda)|}{2} - \binom{|\beta(\lambda)|+1}{2} \\ &= |\lambda| + 2y + 2m - 1. \end{aligned}$$

Thus, we obtain $|\lambda'| > |\lambda|$, violating the assumption that λ has the largest size.

Cases (2), (3), and (4) can be proved similarly. This completes the proof. ∎

Define $E_s = \{3i-1+(s-1)(3t+1) \mid t \le i \le 2t-s-1\}$ for $1 \le s \le t-1$. Then we define the poset $V_{3t-2,3t+1} = (\bigcup_{s=1}^{t-1} E_s) \cup A_0$. Fig. 4 illustrates the Hasse diagram of the poset

$V_{13,16}$. Suppose that $\lambda \in \mathcal{W}_{3t-2,3t+1}$ with $\beta(\lambda) \in \mathcal{T}(P_{3t-2,3t+1})$ and $3(t-1) \notin \beta(\lambda)$. By Lemma 3.6, one can easily check that $\beta(\lambda)$ is a nice order ideal of $\mathcal{O}(V_{3t-2,3t+1})$.

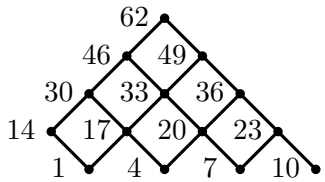


Fig. 4: The Hasse diagram of the poset $V_{13,16}$.

Lemma 3.7 *For an integer $t \geq 2$, let λ be a $(3t-2, 3t+1)$-core partition into distinct parts of the largest size. If $\beta(\lambda) \in \mathcal{T}(P_{3t-2,3t+1})$ and $\beta(\lambda)$ contains an element $x \in E_s$ for $1 \leq s \leq t-1$, then y must also belong to $\beta(\lambda)$ for all $y \in E_s$ with $y > x$.*

Proof. Suppose that λ is a $(3t-2, 3t+1)$-core partition into distinct parts of the largest size and that $\beta(\lambda) \in \mathcal{T}(P_{3t-2,3t+1})$. To prove the lemma, it suffices to show that for each $1 \leq s \leq t-1$, if $\beta(\lambda)$ contains the element $x \in E_s$, then it also contains $x+3 \in E_s$. Assume for contradiction that this property fails. Let s be the minimal index, $1 \leq s \leq t-1$, for which the property does not hold, and x be the smallest element in E_s for which the condition fails. We construct a new order ideal β' of $P_{3t-2,3t+1}$ by replacing each $y \in \beta(\lambda)$ with $y+3$ whenever there exists $p \in E_s$ with $p \leq x$ and $p \leq y$. This construction maintains $|\beta'| = |\beta(\lambda)|$, but strictly increases the sum of elements: $\sum_{x \in \beta'} x > \sum_{x \in \beta(\lambda)} x$, as can be easily verified. Lemma 2.1(1) shows that the partition corresponding to β' has a strictly larger size than λ, contradicting the assumption that λ has the largest size. This completes the proof. ∎

For $1 \leq j \leq t-1$, let $\mu^{(1)}_{t-1,j}$ be the unique order ideal in $\mathcal{T}(P_{3t-2,3t+1})$ that contains $\beta_{1,t-1}$ and the chains consisting of $3t-1, 3t-1+(3t+1), \ldots, 3t-1+(j-1)(3t+1)$. Clearly, $\mu^{(1)}_{t-1,t-1}$ is exactly the set that consists of all elements of $V_{3t-2,3t+1}$. Fig. 4 depicts the order ideal $\mu^{(1)}_{4,4}$, whose elements are shown as solid circles. Let $\delta^{(1)}_{t-1,j}$ be the unique partition such that $\beta(\delta^{(1)}_{t-1,j}) = \mu^{(1)}_{t-1,j}$. By Lemma 2.4, we have $\delta^{(1)}_{t-1,j} \in \mathcal{W}_{3t-2,3t+1}$.

Lemma 3.8 *For an integer $t \geq 2$, let λ be a $(3t-2, 3t+1)$-core partition into distinct parts with $\beta(\lambda) \in \mathcal{T}(P_{3t-2,3t+1})$. If λ has the largest size, then there exists some integer $1 \leq j \leq t-1$ such that $\mu^{(1)}_{t-1,j} \subseteq \beta(\lambda)$.*

Proof. Suppose that λ is a $(3t-2, 3t+1)$-core partition into distinct parts of the largest size with $\beta(\lambda) \in \mathcal{T}(P_{3t-2,3t+1})$. By Lemma 3.7, it suffices to show that $\beta(\lambda)$

contains the element $1+m(3t+1)$ for all $0 \le m \le t-2$. If not, let ℓ be the smallest integer such that $1+\ell(3t+1) \notin \beta(\lambda)$. Choose j to be the largest integer such that $(3t-1)+(j-1)(3t+1) \in \beta(\lambda)$, where $1 \le j \le t-1$. By Lemma 3.7, we have $j \le \ell \le t-2$. Define λ' to be a $(3t-2, 3t+1)$-core partition whose β-set $\beta(\lambda')$ is obtained from $\beta(\lambda)$ by replacing the element $(3t-1)+(j-1)(3t+1)$ with the element $1+\ell(3t+1)$. By Lemma 3.7 and the definition of $\mathcal{T}(P_{3t-2,3t+1})$, we find that $\beta(\lambda')$ is a nice order ideal of $P_{3t-2,3t+1}$ and $\beta(\lambda') \in \mathcal{T}(P_{3t-2,3t+1})$. Moreover, we can verify that $\beta(\lambda')$ maintains the same cardinality as $\beta(\lambda)$ while achieving a greater sum of its elements. Applying Lemma 2.1(1), we conclude that the size of λ' is larger than the size of λ. This contradicts our initial assumption that λ has the largest size. This completes the proof. ∎

Lemma 3.9 *For an integer $t \ge 2$, we have $|\delta^{(1)}_{t-1,j}| \le |\delta^{(1)}_{t-1,t-1}|$ for $1 \le j \le t-1$, with the equality holding if and only if $j = t-1$.*

Proof. By Lemma 2.1(1), we have

$$\begin{aligned}
|\delta^{(1)}_{t-1,j}| &= \sum_{h \in \mu^{(1)}_{t-1,j}} h - \binom{|\mu^{(1)}_{t-1,j}|}{2} \\
&= \sum_{h \in \beta_{1,t-1}} h + \sum_{p=0}^{j-1}(3t-1+p(3t+1)) - \binom{|\beta_{1,t-1}|+j}{2} \\
&= \sum_{h \in \beta_{1,t-1}} h - \binom{|\beta_{1,t-1}|}{2} + \sum_{p=0}^{j-1}(3t-1+p(3t+1)) - \binom{|\beta_{1,t-1}|+j}{2} + \binom{|\beta_{1,t-1}|}{2} \\
&= |\gamma_{1,t-1}| + 3t\binom{j+1}{2} - j(|\beta_{1,t-1}|+1).
\end{aligned}$$

By the definition of $\beta_{1,t-1}$, we have $|\beta_{1,t-1}| = \sum_{j=1}^{t-1} j = \binom{t}{2}$. Hence,

$$|\delta^{(1)}_{t-1,j}| = |\gamma_{1,t-1}| + 3t\binom{j+1}{2} - j\left(\binom{t}{2}+1\right). \tag{3.9}$$

From (3.9), when $j = t-1$, we obtain

$$|\delta^{(1)}_{t-1,t-1}| - |\gamma_{1,t-1}| = 3t\binom{t}{2} - (t-1)\left(\binom{t}{2}+1\right) = \frac{(t-1)(2t^2+t-2)}{2}. \tag{3.10}$$

When $j = 1$, we derive that

$$|\delta^{(1)}_{t-1,1}| - |\gamma_{1,t-1}| = 3t - \left(\binom{t}{2}+1\right) = -\frac{t^2-7t+2}{2}. \tag{3.11}$$

Combining (3.10) and (3.11) above, we get

$$|\delta^{(1)}_{t-1,t-1}| - |\delta^{(1)}_{t-1,1}| = \frac{(t-1)(2t^2+t-2)}{2} + \frac{t^2-7t+2}{2} = (t-2)(t^2+2t-1). \tag{3.12}$$

Since $t \geq 2$, we have $|\delta^{(1)}_{t-1,t-1}| - |\delta^{(1)}_{t-1,1}| \geq 0$, with the equality holding if and only if $t = 2$, at this point, $j = t-1 = 1$.

On the other hand, for fixed t, we observe that $|\delta^{(1)}_{t-1,j}|$ is a quadratic function of j with a positive leading coefficient. Consequently, when j ranges over $[1, t-1]$, the maximum value of $|\delta^{(1)}_{t-1,j}|$ is attained at the endpoint: $j = 1$ or $j = t-1$. By (3.12), we have $|\delta^{(1)}_{t-1,j}| \leq |\delta^{(1)}_{t-1,t-1}|$ for $1 \leq j \leq t-1$, with equality if and only if $j = t-1$. This completes the proof. ∎

Based on the above four lemmas and the definition of $\mu^{(1)}_{t-1,t-1}$, we can directly obtain the following theorem.

Theorem 3.10 *For an integer $t \geq 2$, let λ be a $(3t-2, 3t+1)$-core partition into distinct parts with $\beta(\lambda) \in \mathcal{T}(P_{3t-2,3t+1})$. If λ has the largest size, then $\mu^{(1)}_{t-1,t-1} \subseteq \beta(\lambda)$.*

According to the proof of Lemma 3.6 and the structure of the poset $P_{3t-2,3t+1}$, it remains to determine whether $3(t-1)$ belongs to $\beta(\lambda)$. In what follows, we discuss in detail the existence of $3(t-1)$.

Lemma 3.11 *For an integer $2 \leq t \leq 4$, let λ be a $(3t-2, 3t+1)$-core partition into distinct parts of the largest size. If $\beta(\lambda) \in \mathcal{T}(P_{3t-2,3t+1})$, then $\beta(\lambda) = \mu^{(1)}_{t-1,t-1} \cup \{3(t-1)\}$.*

Proof. Suppose that λ is a $(3t-2, 3t+1)$-core partition into distinct parts of the largest size with $\beta(\lambda) \in \mathcal{T}(P_{3t-2,3t+1})$ for $2 \leq t \leq 4$. To prove this lemma, it suffices to show that $3(t-1) \in \beta(\lambda)$. If not, then by Theorem 3.10, we have $\beta(\lambda) = \mu^{(1)}_{t-1,t-1}$. Define a new partition λ' whose $\beta(\lambda')$ is obtained from $\beta(\lambda)$ by adding the element $3(t-1)$. It is easy to check that λ' is a $(3t-2, 3t+1)$-core partition into distinct parts with $\beta(\lambda') \in \mathcal{T}(P_{3t-2,3t+1})$ and $3(t-1) \in \beta(\lambda')$. Thus,

$$|\beta(\lambda')| = |\beta(\lambda)| + 1 = |\mu^{(1)}_{t-1,t-1}| + 1$$

and

$$\sum_{h \in \beta(\lambda')} h = \sum_{h \in \beta(\lambda)} h + 3(t-1).$$

By Lemma 2.1(1), the size of λ' is given by

$$\begin{aligned}
|\lambda'| &= \sum_{h\in\beta(\lambda')} h - \binom{|\beta(\lambda')|}{2} \\
&= \sum_{h\in\beta(\lambda)} h + 3(t-1) - \binom{|\beta(\lambda)|+1}{2} \\
&= \sum_{h\in\beta(\lambda)} h - \binom{|\beta(\lambda)|}{2} + 3(t-1) - \binom{|\beta(\lambda)|+1}{2} + \binom{|\beta(\lambda)|}{2} \\
&= |\lambda| - |\beta(\lambda)| + 3(t-1).
\end{aligned}$$

By the definition of $\mu^{(1)}_{t-1,t-1}$, we derive

$$|\beta(\lambda)| = |\mu^{(1)}_{t-1,t-1}| = \sum_{j=1}^{t-1} j + (t-1) = (t-1)(\frac{t}{2}+1).$$

Thus,

$$|\lambda'| - |\lambda| = -|\beta(\lambda)| + 3(t-1) = (t-1)(2-\frac{t}{2}). \tag{3.13}$$

Therefore $|\lambda'| \geq |\lambda|$ for $2 \leq t \leq 4$ with equality if and only if $t = 4$, and $|\lambda'| < |\lambda|$ for $t > 4$. This contradicts the assumption that λ has the largest size for $2 \leq t \leq 4$. This completes the proof. ∎

The following result follows immediately from Theorem 3.10 and Lemma 3.11. This theorem is the main result of this subsection.

Theorem 3.12 *For an integer $t \geq 2$, let λ be a $(3t-2, 3t+1)$-core partition into distinct parts with $\beta(\lambda) \in \mathcal{T}(P_{3t-2,3t+1})$. Suppose that λ has the largest size. Then*

(1) if $2 \leq t \leq 4$, then $\beta(\lambda) = \mu^{(1)}_{t-1,t-1} \cup \{3(t-1)\}$,

(2) if $t > 4$, then $\beta(\lambda) = \mu^{(1)}_{t-1,t-1}$.

Moreover, the number of such partitions attaining the largest size is

$$\begin{cases} 2 & \text{if } t = 4, \\ 1 & \text{otherwise.} \end{cases}$$

3.3 The case $\beta(\lambda) \in \mathcal{S}(P_{3t-2,3t+1})$

This subsection studies $(3t-2, 3t+1)$-core partitions λ into distinct parts for which $\beta(\lambda) \in \mathcal{S}(P_{3t-2,3t+1})$. We present four lemmas that characterize $\beta(\lambda)$ when λ has the largest size.

Let $R_0 = \{3i-1 \mid 1 \le i \le t-1\}$. Clearly, $R_0 \subseteq A_1$.

Lemma 3.13 *For an integer $t \ge 2$, let λ be a $(3t-2, 3t+1)$-core partition with distinct parts of the largest size. If $\beta(\lambda) \in \mathcal{S}(P_{3t-2,3t+1})$, then $\beta(\lambda)$ contains all elements in R_0.*

Proof. Let λ be a $(3t-2, 3t+1)$-core partition into distinct parts with the largest size and $\beta(\lambda) \in \mathcal{S}(P_{3t-2,3t+1})$. Suppose, for contradiction, that the lemma fails to hold. Since $3t \in \beta(\lambda)$, it follows from Lemma 2.1 (2) that $2 \in \beta(\lambda)$. Let $3m-1$ be the smallest element of R_0 that is not contained in $\beta(\lambda)$, where $2 \le m \le t-1$. Once m is fixed, let ℓ be the largest integer, with $-1 \le \ell \le t-m-1$, such that the element $3(m+\ell) \in A_2$ of rank 0 is not contained in $\beta(\lambda)$. We now categorize $\beta(\lambda)$ into three classes L_1, L_2, and L_3 as follows.

(i) $L_1 = \{2, 5, \ldots, 3(m-1)-1\}$.

(ii) $L_2 \in \mathcal{O}(G_{3t-2,3t+1})$, where the poset $G_{3t-2,3t+1}$ is given by:

$$G_{3t-2,3t+1} = (\bigcup_{s=1}^{\ell} F_s^{(1)}) \cup (\bigcup_{s=1}^{\ell} F_s^{(2)}) \cup (\bigcup_{s=1}^{\ell+2} F_s^{(3)}).$$

Here, the constituent sets are defined as:

$$F_s^{(1)} = \{3i + (s-1)(3t+1) \mid m \le i \le m+\ell-s\} \quad \text{for } 1 \le s \le \ell,$$
$$F_s^{(2)} = \{3i - 1 + (s-1)(3t+1) \mid m+1 \le i \le m+\ell+1-s\} \quad \text{for } 1 \le s \le \ell,$$
$$F_s^{(3)} = \{3i - 2 + (s-1)(3t+1) \mid m \le i \le m+\ell+2-s\} \quad \text{for } 1 \le s \le \ell+2.$$

(iii) $L_3 \in \mathcal{O}(X_{3t-2,3t+1})$, where $X_{3t-2,3t+1} = \bigcup_{s=1}^{t-\ell-2} F_s^{(4)}$ with

$$F_s^{(4)} = \{3i + (s-1)(3t+1) \mid m+\ell+1 \le i \le m+t-1-s\} \quad \text{for } 1 \le s \le t-\ell-2.$$

Note that $3t \in L_3$.

According to whether $3m-2$ belongs to $\beta(\lambda)$, we divide the analysis into the following two cases: (1) $3m-2 \notin \beta(\lambda)$, (2) $3m-2 \in \beta(\lambda)$. We prove the assertion only for Case (1), as the remaining case follows by a similar argument.

Case (1). $3m-2 \notin \beta(\lambda)$. We proceed to construct a new order ideal β' of $P_{3t-2,3t+1}$ using the following procedure.

- Replace L_1 by $\{2, 5, \ldots, 3(m-1)-1, 3m-1\}$.
- Replace each element x of L_2 by $x+1$.
- Replace each element x of L_3 by $x+3$.

Suppose that the cardinality of L_3 is given by y. Then

$$|\beta'| = |\beta(\lambda)| + 1$$

and

$$\sum_{h\in\beta'} h = \sum_{h\in\beta(\lambda)} h + (3m-1) + 3y + [|\beta(\lambda)| - y - (m-1)] = \sum_{h\in\beta(\lambda)} h + |\beta(\lambda)| + 2y + 2m.$$

Let λ' be the unique partition such that $\beta(\lambda') = \beta'$. By Lemma 2.1(1), the size of λ' is given by

$$\begin{aligned}
|\lambda'| &= \sum_{h\in\beta'} h - \binom{|\beta'|}{2} \\
&= \sum_{h\in\beta(\lambda)} h + |\beta(\lambda)| + 2y + 2m - \binom{|\beta(\lambda)|}{2} + \binom{|\beta(\lambda)|}{2} - \binom{|\beta(\lambda)|+1}{2} \\
&= |\lambda| + 2y + 2m.
\end{aligned}$$

Thus, we obtain $|\lambda'| > |\lambda|$, violating the assumption that λ has the largest size.

Case (2) can be proved similarly. This completes the proof. ∎

Define $R_s = \{3i + (s-1)(3t+1) \mid t \le i \le 2t-1-s\}$ for $1 \le s \le t-1$. It is easily seen that $R_s \subseteq A_{s+1}$ for $1 \le s \le t-1$. Recall that $R_0 = \{3i-1 \mid 1 \le i \le t-1\}$. Then we define the poset $J_{3t-2,3t+1} = \bigcup_{s=0}^{t-1} R_s$. Fig. 5 illustrates the Hasse diagram of the poset $J_{13,16}$. By Lemma 3.13, one can easily check that $\beta(\lambda)$ is a nice order ideal of $\mathcal{O}(J_{3t-2,3t+1})$.

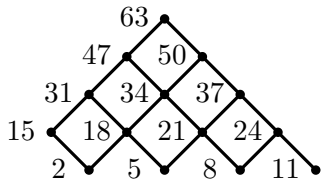


Fig. 5: The Hasse diagram of the poset $J_{13,16}$.

The following lemma can be proved in the same way as Lemma 3.7, so we omit the details.

Lemma 3.14 *For an integer $t \geq 2$, let λ be a $(3t-2, 3t+1)$-core partition with distinct parts that has the largest size. If $\beta(\lambda) \in \mathcal{S}(P_{3t-2,3t+1})$ and $\beta(\lambda)$ contains an element $x \in R_s$ for $1 \leq s \leq t-1$, then y must also belong to $\beta(\lambda)$ for all $y \in R_s$ with $y > x$.*

For $1 \leq j \leq t-1$, define $\mu^{(2)}_{t-1,j}$ as the unique nice order ideal in $\mathcal{S}(P_{3t-2,3t+1})$, which contains $\beta_{2,t-1}$ and the chains consisting of $3t, 3t+(3t+1), \ldots, 3t+(j-1)(3t+1)$. Clearly, $\mu^{(2)}_{t-1,t-1}$ is exactly the set that consists of all elements of $J_{3t-2,3t+1}$. As an illustration, Fig. 5 represents the order ideal $\mu^{(2)}_{4,4}$, with its elements represented by solid circles. Let $\delta^{(2)}_{t-1,j}$ be the unique partition such that $\beta(\delta^{(2)}_{t-1,j}) = \mu^{(2)}_{t-1,j}$. By Lemma 2.4, one can easily check that $\delta^{(2)}_{t-1,j} \in \mathcal{W}_{3t-2,3t+1}$.

Lemma 3.15 *For an integer $t \geq 2$, let λ be a $(3t-2, 3t+1)$-core partition into distinct parts with $\beta(\lambda) \in \mathcal{S}(P_{3t-2,3t+1})$. If λ has the largest size, then there exists some integer $1 \leq j \leq t-1$ such that $\beta(\lambda) = \mu^{(2)}_{t-1,j}$.*

Proof. Suppose that λ is a $(3t-2, 3t+1)$-core partition into distinct parts of the largest size and $\beta(\lambda) \in \mathcal{S}(P_{3t-2,3t+1})$. By Lemma 3.14, it suffices to show that $\beta(\lambda)$ contains the element $2+m(3t+1)$ for all $0 \leq m \leq t-2$. If not, let ℓ be the smallest integer such that $2+\ell(3t+1) \notin \beta(\lambda)$. Moreover, we choose j to be the largest integer that satisfies $3t+(j-1)(3t+1) \in \beta(\lambda)$, where $1 \leq j \leq t-1$. By Lemma 3.14, we have $j \leq \ell \leq t-2$. Let λ' be a $(3t-2, 3t+1)$-core partition, whose corresponding $\beta(\lambda')$ is obtained from $\beta(\lambda)$ by replacing $3t+(j-1)(3t+1)$ with $2+\ell(3t+1)$. By Lemma 3.14 and the definition of $\mathcal{S}(P_{3t-2,3t+1})$, we find that $\beta(\lambda')$ is a nice order ideal of $P_{3t-2,3t+1}$ and $\beta(\lambda') \in \mathcal{S}(P_{3t-2,3t+1})$. Furthermore, we can verify that $\beta(\lambda')$ maintains the same cardinality as $\beta(\lambda)$ while producing a strictly larger sum of its elements. By Lemma 2.1(1), the size of λ' is strictly larger than that of λ, which contradicts the assumption that λ is of the largest size. This completes the proof. ∎

Lemma 3.16 *For $t \geq 2$, we have $|\delta^{(2)}_{t-1,j}| \leq |\delta^{(2)}_{t-1,t-1}|$ for $1 \leq j \leq t-1$, with the equality holding if and only if $j = t-1$.*

Proof. By Lemma 2.1(1), we have

$$\begin{aligned}
|\delta^{(2)}_{t-1,j}| &= \sum_{h\in\mu^{(2)}_{t-1,j}} h - \binom{|\mu^{(2)}_{t-1,j}|}{2} \\
&= \sum_{h\in\beta_{2,t-1}} h + \sum_{p=0}^{j-1}(3t+p(3t+1)) - \binom{|\beta_{2,t-1}|+j}{2} \\
&= \sum_{h\in\beta_{2,t-1}} h - \binom{|\beta_{2,t-1}|}{2} + \sum_{p=0}^{j-1}(3t+p(3t+1)) - \binom{|\beta_{2,t-1}|+j}{2} + \binom{|\beta_{2,t-1}|}{2} \\
&= |\gamma_{2,t-1}| + 3t\binom{j+1}{2} - j|\beta_{2,t-1}|.
\end{aligned}$$

By the definition of $\beta_{2,t-1}$, we have $|\beta_{2,t-1}| = \sum_{j=1}^{t-1} j = \binom{t}{2}$. Hence, we have

$$|\delta^{(2)}_{t-1,j}| = |\gamma_{2,t-1}| + 3t\binom{j+1}{2} - j\binom{t}{2}. \tag{3.14}$$

By (3.14), when $j = t-1$, we derive that

$$|\delta^{(2)}_{t-1,t-1}| - |\gamma_{2,t-1}| = 3t\binom{t}{2} - (t-1)\binom{t}{2} = \frac{t(t-1)(2t+1)}{2}. \tag{3.15}$$

When $j = 1$, we derive that

$$|\delta^{(2)}_{t-1,1}| - |\gamma_{2,t-1}| = 3t - \binom{t}{2} = -\frac{t^2-7t}{2}. \tag{3.16}$$

Combining (3.15) and (3.16) above, we get

$$|\delta^{(2)}_{t-1,t-1}| - |\delta^{(2)}_{t-1,1}| = \frac{t(t-1)(2t+1)}{2} + \frac{t^2-7t}{2} = t(t-2)(t+2). \tag{3.17}$$

Since $t \geq 2$, we have $|\delta^{(2)}_{t-1,t-1}| - |\delta^{(2)}_{t-1,1}| \geq 0$, with the equality holding if and only if $t = 2$, at this point, $j = t-1 = 1$.

On the other hand, for fixed t, we observe that $|\delta^{(2)}_{t-1,j}|$ is a quadratic function of j with a positive leading coefficient. Consequently, when j ranges over $[1, t-1]$, the maximum value of $|\delta^{(2)}_{t-1,j}|$ is attained at the endpoint: $j = 1$ or $j = t-1$. By (3.17), we have $|\delta^{(2)}_{t-1,j}| \leq |\delta^{(2)}_{t-1,t-1}|$ for $1 \leq j \leq t-1$, with equality if and only if $j = t-1$. This completes the proof. ∎

From the four lemmas above and the definition of $\mu^{(2)}_{t-1,t-1}$, we immediately derive the following theorem, which is the key result of this subsection.

Theorem 3.17 *For an integer $t \geq 2$, let λ be a $(3t-2, 3t+1)$-core partition into distinct parts with $\beta(\lambda) \in \mathcal{S}(P_{3t-2,3t+1})$. If λ has the largest size, then $\beta(\lambda) = \mu^{(2)}_{t-1,t-1}$. Moreover, the number of such partitions attaining the largest size is* 1.

The following theorem plays an essential role in the proof of Theorem 1.1.

Theorem 3.18 *For an integer $t \geq 2$, let λ be a $(3t-2, 3t+1)$-core partition into distinct parts. If λ has the largest size, then $\beta(\lambda) = \mu^{(2)}_{t-1,t-1}$.*

Proof. From Theorems 3.5, 3.12, and 3.17, we conclude that

- If $\beta(\lambda) \in \mathcal{M}(P_{3t-2,3t+1})$, then $\beta(\lambda) = \beta_{3,t-1}$.
- If $\beta(\lambda) \in \mathcal{T}(P_{3t-2,3t+1})$, then $\beta(\lambda) = \mu^{(1)}_{t-1,t-1} \cup \{3(t-1)\}$ for $2 \leq t \leq 4$, and $\beta(\lambda) = \mu^{(1)}_{t-1,t-1}$ for $t > 4$.
- If $\beta(\lambda) \in \mathcal{S}(P_{3t-2,3t+1})$, then $\beta(\lambda) = \mu^{(2)}_{t-1,t-1}$.

Recall that $\beta_{3,t-1} = \beta(\gamma_{3,t-1})$, $\mu^{(1)}_{t-1,t-1} = \beta(\delta^{(1)}_{t-1,t-1})$, and $\mu^{(2)}_{t-1,t-1} = \beta(\delta^{(2)}_{t-1,t-1})$. Combining (3.6) and (3.15), we derive that

$$|\delta^{(2)}_{t-1,t-1}| - |\gamma_{3,t-1}| = t^2(t-1). \tag{3.18}$$

Consequently, the inequality $|\delta^{(2)}_{t-1,t-1}| > |\gamma_{3,t-1}|$ holds for all $t \geq 2$. Combining (3.8) and (3.10), we deduce that

$$|\delta^{(1)}_{t-1,t-1}| - |\gamma_{3,t-1}| = \frac{(t-1)(2t^2-t-2)}{2}. \tag{3.19}$$

Since the solutions to the equation $2t^2 - t - 2 = 0$ are $\frac{1\pm\sqrt{17}}{4}$, it follows that $|\delta^{(1)}_{t-1,t-1}| > |\gamma_{3,t-1}|$ for $t \geq 2$. Combining (3.18) and (3.19), we obtain that

$$|\delta^{(2)}_{t-1,t-1}| - |\delta^{(1)}_{t-1,t-1}| = \frac{(t-1)(t+2)}{2}. \tag{3.20}$$

Therefore, we have $|\delta^{(2)}_{t-1,t-1}| > |\delta^{(1)}_{t-1,t-1}|$ for $t \geq 2$.

This reduces the problem to comparing the sizes of $\delta^{(2)}_{t-1,t-1}$ and θ for $2 \leq t \leq 4$, where θ denotes the partition with $\beta(\theta) = \mu^{(1)}_{t-1,t-1} \cup \{3(t-1)\}$. By (3.13), we have

$$|\theta| - |\delta^{(1)}_{t-1,t-1}| = (t-1)(2 - \frac{t}{2}). \tag{3.21}$$

For $2 \leq t \leq 4$, subtracting (3.21) from (3.20) gives

$$|\delta^{(2)}_{t-1,t-1}| - |\theta| = (t-1)^2. \tag{3.22}$$

Hence, we have $|\delta^{(2)}_{t-1,t-1}| > |\theta|$ for $2 \leq t \leq 4$.

Consequently, we observe that $\beta(\lambda) = \mu^{(2)}_{t-1,t-1}$ if λ is of the largest size for $t \geq 2$. This completes the proof. ∎

Having established these preliminary results, we can now prove Theorem 1.1.

Proof of Theorem 1.1. The assertion is clearly satisfied when $t = 1$. For $t \geq 2$, let λ be a $(3t-2, 3t+1)$-core partition into distinct parts of the largest size. According to Theorem 3.18, we have $\lambda = \delta^{(2)}_{t-1,t-1}$. In view of (3.15), we derive that

$$|\delta^{(2)}_{t-1,t-1}| = |\gamma_{2,t-1}| + 3t\binom{t}{2} - (t-1)\binom{t}{2} = |\gamma_{2,t-1}| + \frac{t(t-1)(2t+1)}{2}. \tag{3.23}$$

By Lemma 2.1(1) and the definition of $\beta_{2,t-1}$, we deduce that

$$|\gamma_{2,t-1}| = \sum_{s=1}^{t-1}\sum_{i=1}^{t-s}(3i-1+(s-1)(3t+1)) - \binom{\binom{t}{2}}{2}. \tag{3.24}$$

A straightforward calculation using (3.23) and (3.24) yields that

$$|\lambda| = |\delta^{(2)}_{t-1,t-1}| = \frac{t(t^2-1)(9t+10)}{24},$$

as desired. Moreover, it is easy to check that the number of partitions of the largest size always equals 1. This completes the proof. ∎

4 Proof of Theorem 1.2

This section is devoted to determining the largest size of $(3t-1, 3t+2)$-core partitions with distinct parts. As in previous sections, we construct the poset $P_{3t-1,3t+2}$ as follows. The next lemma can be proved in the same way as Lemma 2.5, and we omit the details.

Lemma 4.1 *Let $t \geq 1$ be an integer. Then we have*

$$P_{3t-1,3t+2} = \bigcup_{s=0}^{3t-1} A^*_s,$$

where $A^*_0 = \{3i-1 \mid 1 \leq i \leq t-1\}$, $A^*_1 = \{3i-2 \mid 1 \leq i \leq 2t-1\}$, and $A^*_s = \{3i+(s-2)(3t+2) \mid 1 \leq i \leq 3t-s\}$ for $2 \leq s \leq 3t-1$.

As an example, Fig. 6 shows the Hasse diagram of the poset $P_{14,17}$. In the poset $P_{3t-1,3t+2}$, the elements of rank 0 are defined to be $1, 2, 3, \ldots, 3t-2$.

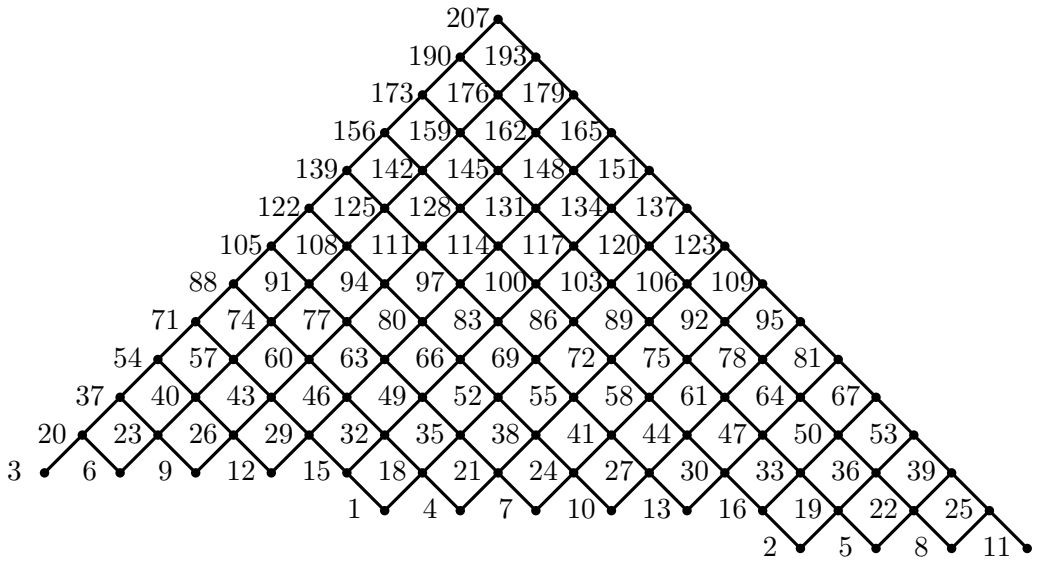


Fig. 6: The Hasse diagram of the poset $P_{14,17}$.

Let $\mathcal{Y}_{3t-1,3t+2}$ denote the set of all $(3t-1, 3t+2)$-core partitions with distinct parts. By Lemma 2.4, if $\lambda \in \mathcal{Y}_{3t-1,3t+2}$, then $\beta(\lambda) \in \mathcal{N}(P_{3t-1,3t+2})$. Note that $3t$ and $3t+1$ cannot both belong to $\beta(\lambda)$. Following the definition of $\mathcal{N}(P_{3t-1,3t+2})$, we also divide this set into three classes:

(i) $\mathcal{E}(P_{3t-1,3t+2})$, the set of all nice order ideals $I \in \mathcal{N}(P_{3t-1,3t+2})$ with $3t+1 \notin I$ and $3t \notin I$,

(ii) $\mathcal{X}(P_{3t-1,3t+2})$, the set of all nice order ideals $I \in \mathcal{N}(P_{3t-1,3t+2})$ with $3t+1 \in I$,

(iii) $\mathcal{K}(P_{3t-1,3t+2})$, the set of all nice order ideals $I \in \mathcal{N}(P_{3t-1,3t+2})$ with $3t \in I$.

Thus,

$$\mathcal{N}(P_{3t-1,3t+2}) = \mathcal{E}(P_{3t-1,3t+2}) \cup \mathcal{X}(P_{3t-1,3t+2}) \cup \mathcal{K}(P_{3t-1,3t+2}).$$

4.1 The case $\beta(\lambda) \in \mathcal{E}(P_{3t-1,3t+2})$

Let t be an integer satisfying $t \geq 2$. For $1 \leq s \leq t$, define

$$F_s^{(1)} = \{3i - 2 + (s-1)(3t+2) \mid 1 \leq i \leq t+1-s\}.$$

For $1 \le s \le t-1$, define

$$F_s^{(2)} = \{3i - 1 + (s-1)(3t+2) \mid 1 \le i \le t-s\}$$

and

$$F_s^{(3)} = \{3i + (s-1)(3t+2) \mid 1 \le i \le t-s\}.$$

We define the poset

$$B^*_{3t-1,3t+2} = (\bigcup_{s=1}^{t} F_s^{(1)}) \cup (\bigcup_{s=1}^{t-1} F_s^{(2)}) \cup (\bigcup_{s=1}^{t-1} F_s^{(3)}).$$

Then

$$\mathcal{E}(P_{3t-1,3t+2}) = \mathcal{N}(B^*_{3t-1,3t+2}).$$

Hence, we construct the Hasse diagram of the poset $B^*_{3t-1,3t+2}$. Fig. 7 illustrates the Hasse diagram of the poset $B^*_{14,17}$. In Fig. 7, from right to left are $\bigcup_{s=1}^{5} F_s^{(1)}$, $\bigcup_{s=1}^{4} F_s^{(2)}$, and $\bigcup_{s=1}^{4} F_s^{(3)}$, respectively.

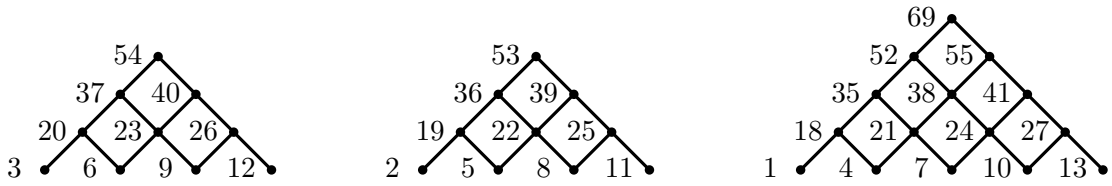


Fig. 7: The Hasse diagram of the poset $B^*_{14,17}$.

Recall that U_t denotes the unique order ideal consisting of all elements of $P_{t,t+1}$. For $1 \le i \le 3t-2$, let $\alpha_{i,0}$ be the unique order ideal in $\mathcal{O}(B^*_{3t-1,3t+2})$ consisting of the elements $i+3, i+6, \ldots, i+3(t-1-\lfloor \frac{i+1}{3} \rfloor)$ and isomorphic to $U_{t-\lfloor \frac{i+1}{3} \rfloor}$. Note that $\alpha_{i,0} = \varnothing$ for $i \in \{3(t-1)-1, 3(t-1), 3t-2\}$. Clearly, $\alpha_{i,0} \in \mathcal{N}(P_{3t-1,3t+2})$.

For $1 \le i \le 3t-2$ and $1 \le j \le t - \lfloor \frac{i+1}{3} \rfloor$, define $\alpha_{i,j}$ to be the unique order ideal in $\mathcal{O}(B^*_{3t-1,3t+2})$, which contains $\alpha_{i,0}$ and the chain consisting of $i, i+(3t+2), \ldots, i+(j-1)(3t+2)$. In particular, $\alpha_{3t-2,1} = \{3t-2\}$, $\alpha_{3(t-1),1} = \{3(t-1)\}$, and $\alpha_{3(t-1)-1,1} = \{3(t-1)-1\}$. Clearly, $\alpha_{1,t}$ (resp. $\alpha_{2,t-1}$, $\alpha_{3,t-1}$) is exactly the set consisting of all elements of $\bigcup_{s=1}^{t} F_s^{(1)}$ (resp. $\bigcup_{s=1}^{t-1} F_s^{(2)}$, $\bigcup_{s=1}^{t-1} F_s^{(3)}$). Fig. 7 illustrates the order ideals $\alpha_{1,5}$, $\alpha_{2,4}$, and $\alpha_{3,4}$ from right to left, with elements indicated by solid circles. Clearly, $\alpha_{i,j} \in \mathcal{N}(P_{3t-1,3t+2})$. Let $\eta_{i,j}$ be the unique partition such that $\beta(\eta_{i,j}) = \alpha_{i,j}$. By Lemma 2.4, we have $\eta_{i,j} \in \mathcal{Y}_{3t-1,3t+2}$.

To characterize the nice order ideals in $\mathcal{N}(B^*_{3t-1,3t+2})$ corresponding to the largest $(3t-1, 3t+2)$-core partitions into distinct parts, we require the following three lemmas. The proofs are similar to those of Lemmas 3.2-3.4 and are therefore omitted.

Lemma 4.2 *For an integer $t \geq 2$, let λ be a $(3t-1, 3t+2)$-core partition into distinct parts of the largest size with $\beta(\lambda) \in \mathcal{E}(P_{3t-1,3t+2})$. If $\beta(\lambda)$ contains an element $x \in F_s^{(i)}$ for some admissible s and $i \in \{1,2,3\}$, then it contains all elements $y \in F_s^{(i)}$ with $y > x$.*

Lemma 4.3 *For an integer $t \geq 2$, let λ be a $(3t-1, 3t+2)$-core partition into distinct parts with $\beta(\lambda) \in \mathcal{E}(P_{3t-1,3t+2})$. If λ has the largest size, then there exist some integers $1 \leq i \leq 3t-2$ and $1 \leq j \leq t - \lfloor \frac{i+1}{3} \rfloor$ such that $\beta(\lambda) = \alpha_{i,j}$.*

Lemma 4.4 *For $1 \leq i \leq 3t-2$, we have $|\eta_{i,j}| \leq |\eta_{i,t-\lfloor \frac{i+1}{3} \rfloor}|$ for $1 \leq j \leq t - \lfloor \frac{i+1}{3} \rfloor$, with the equality holding if and only if $j = t - \lfloor \frac{i+1}{3} \rfloor$. Moreover, when $1 \leq i \leq 3(t-1)-2$ and $t \geq 3$, we have $|\eta_{i,t-\lfloor \frac{i+1}{3} \rfloor}| > |\eta_{i+3,t-\lfloor \frac{i+1}{3} \rfloor - 1}|$.*

The following theorem gives a characterization of the largest $(3t-1, 3t+2)$-core partition with distinct parts whose β-set is contained in $\mathcal{E}(P_{3t-1,3t+2})$.

Theorem 4.5 *For an integer $t \geq 2$, let λ be a $(3t-1, 3t+2)$-core partition into distinct parts with $\beta(\lambda) \in \mathcal{E}(P_{3t-1,3t+2})$. If λ has the largest size, then $\beta(\lambda) = \alpha_{1,t}$. Moreover, the number of such partitions attaining the largest size is 1.*

Proof. Suppose that λ is a $(3t-1, 3t+2)$-core partition into distinct parts with the largest size and $\beta(\lambda) \in \mathcal{E}(P_{3t-1,3t+2})$. By Lemmas 4.3 and 4.4, we have $\beta(\lambda) \in \{\alpha_{1,t}, \alpha_{2,t-1}, \alpha_{3,t-1}\}$. By the definition of $\alpha_{1,t}$, $\alpha_{2,t-1}$, and $\alpha_{3,t-1}$, we have

$$|\alpha_{1,t}| - t = |\alpha_{2,t-1}| = |\alpha_{3,t-1}| = \frac{t(t-1)}{2} \tag{4.1}$$

and

$$\sum_{h \in \alpha_{3,t-1}} h = \sum_{h \in \alpha_{2,t-1}} h + \frac{t(t-1)}{2}.$$

By Lemma 2.1(1), we obtain that

$$|\eta_{3,t-1}| - |\eta_{2,t-1}| = \frac{t(t-1)}{2},$$

and hence $|\eta_{3,t-1}| > |\eta_{2,t-1}|$ for $t \geq 2$. Moreover,

$$\sum_{h \in \alpha_{1,t}} h = \sum_{h \in \alpha_{3,t-1}} h - t(t-1) + \sum_{p=0}^{t-1} [(3t-2) + p(3t-1)] = \sum_{h \in \alpha_{3,t-1}} h + \frac{t(3t^2-1)}{2}. \tag{4.2}$$

Combining Lemma 2.1(1) with equations (4.1) and (4.2), we obtain that

$$\begin{aligned}
|\eta_{1,t}| &= \sum_{h\in\alpha_{1,t}} h - \binom{|\alpha_{1,t}|}{2} \\
&= \sum_{h\in\alpha_{1,t}} h - \binom{|\alpha_{3,t-1}|+t}{2} \\
&= \sum_{h\in\alpha_{3,t-1}} h + \frac{t(3t^2-1)}{2} - \binom{|\alpha_{3,t-1}|+t}{2} \\
&= \sum_{h\in\alpha_{3,t-1}} h - \binom{|\alpha_{3,t-1}|}{2} + \frac{t(3t^2-1)}{2} + \binom{|\alpha_{3,t-1}|}{2} - \binom{|\alpha_{3,t-1}|+t}{2} \\
&= |\eta_{3,t-1}| + \frac{t(3t^2-1)}{2} + \binom{|\alpha_{3,t-1}|}{2} - \binom{|\alpha_{3,t-1}|+t}{2} \\
&= |\eta_{3,t-1}| + \frac{t(3t^2-1)}{2} - |\alpha_{3,t-1}|\cdot t - \frac{t(t-1)}{2} \\
&= |\eta_{3,t-1}| + \frac{t(3t^2-1)}{2} - \frac{t(t-1)}{2}\cdot t - \frac{t(t-1)}{2} \\
&= |\eta_{3,t-1}| + t^3.
\end{aligned}$$

Since $t \geq 2$, we have

$$|\eta_{1,t}| - |\eta_{3,t-1}| = t^3 > 0. \tag{4.3}$$

Therefore, $\lambda = \eta_{1,t}$ and $\beta(\lambda) = \alpha_{1,t}$ as desired. Moreover, one easily checks that this largest partition is unique, completing the proof. ∎

4.2 The case $\beta(\lambda) \in \mathcal{X}(P_{3t-1,3t+2})$

To characterize the nice order ideals in $\mathcal{X}(P_{3t-1,3t+2})$ corresponding to the largest $(3t-1,3t+2)$-core partitions into distinct parts, we require the following four lemmas. The proofs are similar to those of Lemmas 3.6-3.9 and are therefore omitted.

Lemma 4.6 *For an integer $t \geq 2$, let λ be a $(3t-1,3t+2)$-core partition into distinct parts of the largest size. If $\beta(\lambda) \in \mathcal{X}(P_{3t-1,3t+2})$, then $\beta(\lambda)$ contains all elements of A_0^*.*

For $1 \leq s \leq t-1$, define $Z_s = \{3i-2+(s-1)(3t+2) \mid t+1 \leq i \leq 2t-s\}$. Set $V^*_{3t-1,3t+2} = (\bigcup_{s=1}^{t-1} Z_s) \cup A_0^*$. Fig. 8 shows the Hasse diagram of the poset $V^*_{14,17}$. Suppose that $\lambda \in \mathcal{Y}_{3t-1,3t+2}$ with $\beta(\lambda) \in \mathcal{X}(P_{3t-1,3t+2})$. By Lemma 4.6, $\beta(\lambda)$ is a nice order ideal of $\mathcal{O}(V^*_{3t-1,3t+2})$.

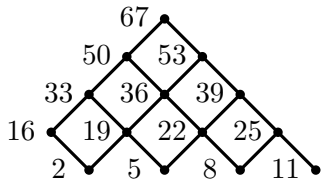


Fig. 8: The Hasse diagram of the poset $V^*_{14,17}$.

Lemma 4.7 *For an integer $t \geq 2$, let λ be a $(3t-1, 3t+2)$-core partition into distinct parts of the largest size. If $\beta(\lambda) \in \mathcal{X}(P_{3t-1,3t+2})$ and $\beta(\lambda)$ contains an element $x \in Z_s$ for $1 \leq s \leq t-1$, then y must also belong to $\beta(\lambda)$ for all $y \in Z_s$ with $y > x$.*

For $1 \leq j \leq t-1$, let $\rho^{(1)}_{t-1,j}$ be the unique order ideal in $\mathcal{X}(P_{3t-1,3t+2})$ that contains $\alpha_{2,t-1}$ and the chains consisting of $3t+1, 3t+1+(3t+2), \ldots, 3t+1+(j-1)(3t+2)$. Clearly, $\rho^{(1)}_{t-1,t-1}$ is exactly the set that consists of all elements of $V^*_{3t-1,3t+2}$. Fig. 8 depicts the order ideal $\rho^{(1)}_{4,4}$, whose elements are shown as solid circles. Let $\xi^{(1)}_{t-1,j}$ be the unique partition such that $\beta(\xi^{(1)}_{t-1,j}) = \rho^{(1)}_{t-1,j}$. By Lemma 2.4, we have $\xi^{(1)}_{t-1,j} \in \mathcal{Y}_{3t-1,3t+2}$.

Lemma 4.8 *For an integer $t \geq 2$, let λ be a $(3t-1, 3t+2)$-core partition into distinct parts with $\beta(\lambda) \in \mathcal{X}(P_{3t-1,3t+2})$. If λ has the largest size, then there exists some integer $1 \leq j \leq t-1$ such that $\rho^{(1)}_{t-1,j} \subseteq \beta(\lambda)$.*

Lemma 4.9 *For an integer $t \geq 2$, we have $|\xi^{(1)}_{t-1,j}| \leq |\xi^{(1)}_{t-1,t-1}|$ for $1 \leq j \leq t-1$, with the equality holding if and only if $j = t-1$.*

For $1 \leq s \leq t$, define $Z^*_s = \{3i-2+(s-1)(3t+2) \mid t \leq i \leq 2t-s\}$. Set $W_{3t-1,3t+2} = (\bigcup_{s=1}^{t} Z^*_s) \cup A^*_0$. Fig. 9 shows the Hasse diagram of the poset $W_{14,17}$. For $1 \leq j \leq t$, let $\rho^{(2)}_{t-1,j}$ be the unique order ideal in $\mathcal{X}(P_{3t-1,3t+2})$ that contains $\rho^{(1)}_{t-1,t-1}$ and the chains consisting of $3t-2, 3t-2+(3t+2), \ldots, 3t-2+(j-1)(3t+2)$. Then $\rho^{(2)}_{t-1,t}$ is exactly the set that consists of all elements of $W_{3t-1,3t+2}$. Fig. 9 depicts the order ideal $\rho^{(2)}_{4,5}$, whose elements are shown as solid circles. Let $\xi^{(2)}_{t-1,j}$ be the unique partition such that $\beta(\xi^{(2)}_{t-1,j}) = \rho^{(2)}_{t-1,j}$. By Lemma 2.4, we have $\xi^{(2)}_{t-1,j} \in \mathcal{Y}_{3t-1,3t+2}$.

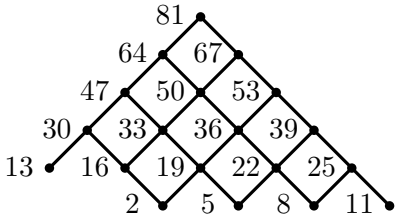


Fig. 9: The Hasse diagram of the poset $W_{14,17}$.

Lemma 4.10 *For an integer $t \geq 2$, we have $|\xi^{(2)}_{t-1,j}| \leq |\xi^{(2)}_{t-1,t}|$ for $1 \leq j \leq t$, and $|\xi^{(1)}_{t-1,t-1}| \leq |\xi^{(2)}_{t-1,t}|$, with the equality holding if and only if $j = t$.*

Proof. By Lemma 2.1(1), we have

$$\begin{aligned}
|\xi^{(2)}_{t-1,j}| &= \sum_{h\in\rho^{(2)}_{t-1,j}} h - \binom{|\rho^{(2)}_{t-1,j}|}{2} \\
&= \sum_{h\in\rho^{(1)}_{t-1,t-1}} h + \sum_{p=0}^{j-1}[(3t-2)+p(3t+2)] - \binom{|\rho^{(1)}_{t-1,t-1}|+j}{2} \\
&= \sum_{h\in\rho^{(1)}_{t-1,t-1}} h - \binom{|\rho^{(1)}_{t-1,t-1}|}{2} + \sum_{p=0}^{j-1}[(3t-2)+p(3t+2)] - \binom{|\rho^{(1)}_{t-1,t-1}|+j}{2} + \binom{|\rho^{(1)}_{t-1,t-1}|}{2} \\
&= |\xi^{(1)}_{t-1,t-1}| + 3t\binom{j+1}{2} + \frac{j(j-5)}{2} - j|\rho^{(1)}_{t-1,t-1}|.
\end{aligned}$$

By the definition of $\rho^{(1)}_{t-1,t-1}$, we have $|\rho^{(1)}_{t-1,t-1}| = (t-1) + \sum_{j=1}^{t-1} j = \frac{(t-1)(t+2)}{2}$. Hence, we have

$$|\xi^{(2)}_{t-1,j}| = |\xi^{(1)}_{t-1,t-1}| + 3t\binom{j+1}{2} + \frac{j(j-5)}{2} - j\frac{(t-1)(t+2)}{2}. \tag{4.4}$$

By (4.4), when $j = t$, we derive that

$$|\xi^{(2)}_{t-1,t}| - |\xi^{(1)}_{t-1,t-1}| = 3t\binom{t+1}{2} + \frac{t(t-5)}{2} - \frac{t(t-1)(t+2)}{2} = t^3 + \frac{3}{2}t^2 - \frac{3}{2}t. \tag{4.5}$$

Since $t \geq 2$, we have $|\xi^{(2)}_{t-1,t}| > |\xi^{(1)}_{t-1,t-1}|$ as desired. When $j = 1$, we derive that

$$|\xi^{(2)}_{t-1,1}| - |\xi^{(1)}_{t-1,t-1}| = 3t - 2 - \frac{(t-1)(t+2)}{2} = -\frac{1}{2}t^2 + \frac{5}{2}t - 1. \tag{4.6}$$

Combining (4.5) and (4.6) above, we get

$$|\xi^{(2)}_{t-1,t}| - |\xi^{(2)}_{t-1,1}| = (t-1)(t^2+3t-1). \tag{4.7}$$

Since $t \geq 2$, we have $|\xi^{(2)}_{t-1,t}| - |\xi^{(2)}_{t-1,t-1}| > 0$.

For fixed t, we observe that $|\xi^{(2)}_{t-1,j}|$ is a quadratic function of j with a positive leading coefficient. Consequently, when j ranges over $[1,t]$, the maximum value of $|\xi^{(2)}_{t-1,j}|$ is attained at the endpoint: $j = 1$ or $j = t$. By (4.7), we have $|\xi^{(2)}_{t-1,j}| \leq |\xi^{(2)}_{t-1,t}|$ for $1 \leq j \leq t$, with equality if and only if $j = t$. This completes the proof. ∎

The following theorem follows immediately from Lemmas 4.9 and 4.10.

Theorem 4.11 *For an integer $t \geq 2$, let λ be a $(3t-1, 3t+2)$-core partition into distinct parts with $\beta(\lambda) \in \mathcal{X}(P_{3t-1,3t+2})$. If λ has the largest size, then $\beta(\lambda) = \rho^{(2)}_{t-1,t}$.*

4.3 The case $\beta(\lambda) \in \mathcal{K}(P_{3t-1,3t+2})$

The following four lemmas characterize the nice order ideals in $\mathcal{K}(P_{3t-1,3t+2})$ associated with the largest $(3t-1, 3t+2)$-core partitions into distinct parts; their proofs are analogous to those of Lemmas 3.13–3.16 and are omitted.

Let $R^*_0 = \{3i-2 \mid 1 \leq i \leq t\}$. Clearly, $R^*_0 \subseteq A^*_1$.

Lemma 4.12 *For an integer $t \geq 2$, let λ be a $(3t-1, 3t+2)$-core partition into distinct parts of the largest size. If $\beta(\lambda) \in \mathcal{K}(P_{3t-1,3t+2})$, then $\beta(\lambda)$ contains all elements of R^*_0.*

For $1 \leq s \leq t$, define $R^*_s = \{3i + (s-1)(3t+2) \mid t \leq i \leq 2t-s\}$. Set $J^*_{3t-1,3t+2} = (\bigcup_{s=0}^{t} R^*_s)$. Fig. 10 shows the Hasse diagram of the poset $J^*_{14,17}$. Suppose that $\lambda \in \mathcal{Y}_{3t-1,3t+2}$ with $\beta(\lambda) \in \mathcal{K}(P_{3t-1,3t+2})$. By Lemma 4.12, $\beta(\lambda)$ is a nice order ideal of $\mathcal{O}(J^*_{3t-1,3t+2})$.

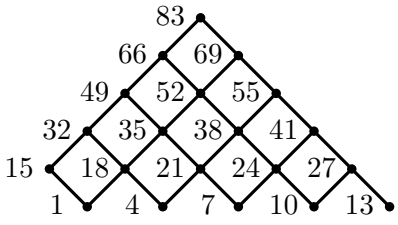


Fig. 10: The Hasse diagram of the poset $J^*_{14,17}$.

Lemma 4.13 *For an integer $t \geq 2$, let λ be a $(3t-1, 3t+2)$-core partition into distinct parts of the largest size. If $\beta(\lambda) \in \mathcal{K}(P_{3t-1,3t+2})$ and $\beta(\lambda)$ contains an element $x \in R_s^*$ for $1 \leq s \leq t$, then y must also belong to $\beta(\lambda)$ for all $y \in R_s^*$ with $y > x$.*

For $1 \leq j \leq k \leq t$, let $\varrho_{k,j}$ be the unique order ideal in $\mathcal{K}(P_{3t-1,3t+2})$ that contains $\alpha_{1,k}$ and the chains consisting of $3t, 3t+(3t+2), \ldots, 3t+(j-1)(3t+2)$. Clearly, $\varrho_{t,t}$ is exactly the set that consists of all elements of $J^*_{3t-1,3t+2}$. Fig. 10 depicts the order ideal $\varrho_{5,5}$, whose elements are shown as solid circles. Let $\zeta_{k,j}$ be the unique partition such that $\beta(\zeta_{k,j}) = \varrho_{k,j}$. By Lemma 2.4, we have $\zeta_{k,j} \in \mathcal{Y}_{3t-1,3t+2}$.

Lemma 4.14 *For an integer $t \geq 2$, let λ be a $(3t-1, 3t+2)$-core partition into distinct parts with $\beta(\lambda) \in \mathcal{K}(P_{3t-1,3t+2})$. If λ has the largest size, then there exists some integer $1 \leq j \leq t$ such that $\varrho_{t,j} \subseteq \beta(\lambda)$.*

Lemma 4.15 *For an integer $t \geq 2$, we have $|\zeta_{t,j}| \leq |\zeta_{t,t}|$ for $1 \leq j \leq t$, with the equality holding if and only if $j = t$.*

Based on the four lemmas above, we immediately obtain the following result.

Theorem 4.16 *For an integer $t \geq 2$, let λ be a $(3t-1, 3t+2)$-core partition into distinct parts with $\beta(\lambda) \in \mathcal{K}(P_{3t-1,3t+2})$. If λ has the largest size, then $\beta(\lambda) = \varrho_{t,t}$.*

The following theorem plays an essential role in the proof of Theorem 1.2.

Theorem 4.17 *For an integer $t \geq 2$, let λ be a $(3t-1, 3t+2)$-core partition into distinct parts. If λ has the largest size, then $\beta(\lambda) = \varrho_{t,t}$.*

Proof. By Theorems 4.5, 4.11, and 4.16, we conclude that

- If $\beta(\lambda) \in \mathcal{E}(P_{3t-1,3t+2})$, then $\beta(\lambda) = \alpha_{1,t}$.
- If $\beta(\lambda) \in \mathcal{X}(P_{3t-1,3t+2})$, then $\beta(\lambda) = \rho^{(2)}_{t-1,t}$.
- If $\beta(\lambda) \in \mathcal{K}(P_{3t-1,3t+2})$, then $\beta(\lambda) = \varrho_{t,t}$.

Recall that $\alpha_{1,t} = \beta(\eta_{1,t})$, $\rho^{(2)}_{t-1,t} = \beta(\xi^{(2)}_{t-1,t})$, and $\varrho_{t,t} = \beta(\zeta_{t,t})$. By Lemma 2.1(1), we have

$$\begin{aligned}
|\zeta_{t,t}| &= \sum_{h\in\varrho_{t,t}} h - \binom{|\varrho_{t,t}|}{2} \\
&= \sum_{h\in\alpha_{1,t}} h + \sum_{p=0}^{t-1}[3t + p(3t+2)] - \binom{|\alpha_{1,t}| + t}{2} \\
&= \sum_{h\in\alpha_{1,t}} h - \binom{|\alpha_{1,t}|}{2} + \sum_{p=0}^{t-1}[3t + p(3t+2)] - \binom{|\alpha_{1,t}| + t}{2} + \binom{|\alpha_{1,t}|}{2} \\
&= |\eta_{1,t}| + \frac{t(2t^2+3t-1)}{2}.
\end{aligned}$$

Since $t \geq 2$, we have

$$|\zeta_{t,t}| - |\eta_{1,t}| = \frac{t(2t^2+3t-1)}{2} > 0. \tag{4.8}$$

Thus, $|\zeta_{t,t}| > |\eta_{1,t}|$. Moreover, by the definition of $\varrho_{k,j}$, we have

$$\begin{aligned}
|\zeta_{t,t}| &= \sum_{h\in\varrho_{t,t}} h - \binom{|\varrho_{t,t}|}{2} \\
&= \sum_{h\in\varrho_{t-1,t-1}} h + \sum_{p=0}^{t}[(3t-2) + p(3t-1)] - \binom{|\varrho_{t-1,t-1}| + (t+1)}{2} \\
&= \sum_{h\in\varrho_{t-1,t-1}} h - \binom{|\varrho_{t-1,t-1}|}{2} + \sum_{p=0}^{t}[(3t-2) + p(3t-1)] - \binom{|\varrho_{t-1,t-1}| + (t+1)}{2} + \binom{|\varrho_{t-1,t-1}|}{2} \\
&= |\zeta_{t-1,t-1}| + \frac{(3t-2)(t+2)(t+1)}{2} - (t+1)|\varrho_{t-1,t-1}|.
\end{aligned}$$

Since $|\varrho_{t-1,t-1}| = \frac{(t+2)(t-1)}{2}$, we have

$$|\zeta_{t,t}| - |\zeta_{t-1,t-1}| = \frac{(3t-2)(t+2)(t+1)}{2} - \frac{(t+2)(t-1)(t+1)}{2} = \frac{(2t-1)(t+2)(t+1)}{2}. \tag{4.9}$$

By construction, $\rho^{(1)}_{t-1,t-1}$ and $\varrho_{t-1,t-1}$ are isomorphic and have the same cardinality. Indeed, $\rho^{(1)}_{t-1,t-1}$ is obtained from $\varrho_{t-1,t-1}$ by replacing each element x with $x+1$. Consequently,

$$\sum_{h\in\rho^{(1)}_{t-1,t-1}} h = \sum_{h\in\varrho_{t-1,t-1}} h + |\varrho_{t-1,t-1}|. \tag{4.10}$$

By the definition of $\rho^{(2)}_{t-1,t}$, we have

$$|\rho^{(2)}_{t-1,t}| = |\rho^{(1)}_{t-1,t-1}| + t \tag{4.11}$$

and

$$\sum_{h\in\rho^{(2)}_{t-1,t}} h = \sum_{h\in\rho^{(1)}_{t-1,t-1}} h + \sum_{p=0}^{t-1}[(3t-2)+p(3t+2)]. \tag{4.12}$$

Combining (4.10), (4.11), and (4.12), we derive that

$$\begin{aligned}
|\xi^{(2)}_{t-1,t}| &= \sum_{h\in\rho^{(2)}_{t-1,t}} h - \binom{|\rho^{(2)}_{t-1,t}|}{2} \\
&= \sum_{h\in\rho^{(1)}_{t-1,t-1}} h + \sum_{p=0}^{t-1}[(3t-2)+p(3t+2)] - \binom{|\rho^{(1)}_{t-1,t-1}|+t}{2} \\
&= \sum_{h\in\varrho_{t-1,t-1}} h + |\varrho_{t-1,t-1}| + \sum_{p=0}^{t-1}[(3t-2)+p(3t+2)] - \binom{|\varrho_{t-1,t-1}|+t}{2} \\
&= |\zeta_{t-1,t-1}| + |\varrho_{t-1,t-1}| + \sum_{p=0}^{t}[(3t-2)+p(3t+2)] - |\varrho_{t-1,t-1}|t - \frac{t(t-1)}{2} \\
&= |\zeta_{t-1,t-1}| - |\varrho_{t-1,t-1}|(t-1) + \frac{(3t^2+4t-5)t}{2}.
\end{aligned}$$

Since $|\varrho_{t-1,t-1}| = \frac{(t+2)(t-1)}{2}$, we have

$$|\xi^{(2)}_{t-1,t}| - |\zeta_{t-1,t-1}| = -\frac{(t+2)(t-1)^2}{2} + \frac{(3t^2+4t-5)t}{2} = t^3+2t^2-t-1. \tag{4.13}$$

Combining (4.9) and (4.13), we obtain that

$$|\zeta_{t,t}| - |\xi^{(2)}_{t-1,t}| = \frac{(2t-1)(t+2)(t+1)}{2} - (t^3+2t^2-t-1) = \frac{t(t+3)}{2} > 0,$$

and therefore $|\zeta_{t,t}| > |\xi^{(2)}_{t-1,t}|$ for $t \geq 2$. This completes the proof. ∎

We now prove Theorem 1.2.

Proof of Theorem 1.2. The assertion is clear for $t = 1$. Assume $t \geq 2$, and let λ be a $(3t-1, 3t+2)$-core partition into distinct parts of the largest size. By Theorem 4.17, we have $\lambda = \zeta_{t,t}$. By Lemma 2.1(1) and the definition of $\alpha_{1,t}$, we deduce that

$$|\eta_{1,t}| = \sum_{s=1}^{t}\sum_{i=1}^{t+1-s}[3i-2+(s-1)(3t+2)] - \binom{\binom{t+1}{2}}{2}. \tag{4.14}$$

In view of (4.8), we obtain

$$|\zeta_{t,t}| = |\eta_{1,t}| + \frac{t(2t^2+3t-1)}{2}. \tag{4.15}$$

A direct computation using (4.14) and (4.15) yields

$$|\lambda| = |\zeta_{t,t}| = \frac{(9t^3 + 38t^2 + 39t - 14)t}{24}.$$

Moreover, the largest partition is unique. This completes the proof. ∎

Acknowledgments

The authors thank the anonymous referees for their valuable comments and suggestions, which greatly improved the quality of the paper. This work was supported by the National Science Foundation of China [Grant No. 12201155].

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